# *2025 Spring Math*

## HTOP,PDE,COMPLEX,ALGO

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# Contents

Chapter 1

# Functions of Complex Variables

## §1.1 Lecture 1 (02-03)Complex numbers

### §1.1.1 Introduction

You know the sets

$$(R, +, \times) and (Q, +, \times)$$

$Q(\sqrt{2}) = \{a + b\sqrt{2} | a, b \in Q\}$ $Q(\sqrt{2})$, they are infinite sets.

**Definition 1.1.1: Fields**

We say that a set $F$ together with two binary operations, called addition $+$,$F \times F \to F$, and multiplication $\times$, $F \times F \to F$ satisfying:

- associativity $(a + b) + c = a + (b + c)$
- commutativity $a + b = b + a, ab = ba$
- existence of identities
  there is a $O \in F$ such that $a + O = a$ for all $a \in F$
  there is a $1 \in F$ such that $a \times 1 = a$ for all $a \in F$
- existence of inverses
  for every $a \in F$ there is a $b \in F$ such that $a + b = O$
  for every $a \in F$ there is a $b \in F$ such that $a \times b = 1$
- distributivity
  $a \times (b + c) = a \times b + a \times c$

**Example 1: Fields**

- For every prime number $p$, and $n \geq 0$,$n \in N$, there is a field $(F_{p^n}, +, \times)$ with $p^n = q$ elements (cardinality $q$).
- (C,+,x) is a field

Question: $(R^d, +)$,$d \geq 2$, is there a multiplication, $\times$ such that $(R^d, +, x)$ is a field?

- $R^2$ , sclar product, $x \cdot y = x_1 y_1 + x_2 y_2 \in R$ is not a field because $R^2 \times R^2 \to R$.
- cross product, not satisfying Definition 1 and 5.
- d=2, there is a good defination of multiplication, $(R^2, +, \times) \cong (C, +, \times)$.

**Lemma 1.1.1**

Consider the vector space $(R^2, +)$
Define the multiplication: $(a, b) \times (c, d) = (ac - bd, ad + bc)$
Then $(R^2, +, \times)$ is a field.

Proof:

- You can check that $\times$ is commutative and satisfies associativity and distributivity.
- We can check $(1, 0)$ is a multiplication identity.
- also if $(a, b) \neq (0, 0)$, then

$$(a, b) \times (\frac{a}{a^2 + b^2}, \frac{-b}{a^2 + b^2}) = (\frac{a^2 + b^2}{a^2 + b^2}, \frac{-ab + ab}{a^2 + b^2}) = (1, 0)$$

# §1.2 Lecture 2 (02-05)Complex numbers

## §1.2.1 The algebra of complex numbers

It is important to classify sets:
in Set Theory: bijection
in Topology: homeomorphism
For **Fields**: isomorphism

**Definition 1.2.1: isomorphism**

We say that fields $F$ and $F'$ are isomorphic if there exists a funtion

$$i : F \to F'$$

which is a bijection and preserve the binary operations

$$i(a + b) = i(a) + i(b)$$

and

$$i(ab) = i(a)i(b)$$

.
i is called an isomorphism and we write $F \cong F'$

**Comments:**
There is a unque field up to isomorphism of order (its cardinality)

$$p^n : F_{p^n}$$

**Definition 1.2.2: subfield**

Given a field (F,+,x), we say that (E,+,x) is a subfield of (F,+,x) if E $\subset$ F and the addition of multiplication of E is the same addition and multiplication of F.

**Example 1: subfield**

$$Q \subset R$$
$$Q \subset Q(\sqrt{2}) \subset R \subset R^2$$

**Definition 1.2.3: complex number**

Let F be a field such that:

- R is a subfield of F
- there is an element $i \in F$ which is the root of the equation

$$x^2 + 1 = 0(i^2 = -1)$$

Then we define the field of complex numbers $C$ as

$$C = \{x \in F : x = \alpha + i\beta, \alpha, \beta \in R\}$$

**Comments:**

- We need to show that there is at least one field $F$ satisfying these properties. **Actually,** $(F, +, \times) = (R, +, \times)$ **satisfies these properties.**
- $R' = \{x \in R : x = (\alpha, 0), \alpha \in R\} \cong R \rightarrow$ R is a subfield of $R^2$ .
- Define $i := (0, 1) \in R^2$

$$(a, b) \times (c, d) := (ac - bd, ad + bc)$$

Then,

$$i^2 = (0, 1)^2 = (0, 1) \times (0, 1) = (-1, 0)$$

**Example 2: exercise**

Set of $2 \times 2$ real matrices of the form

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}, a, b \in R$$

This set together with the standard addition and multiplication of matrices is a field and it is isomorphic to $(R^2, +, \times)$ .
$F = R^2, C = R^2$

**Comments:**
It turns out that any defination of $C$ is unique (up to isomorphisms).

**Lemma 1.2.1**

Let $F$ and $F'$ be two fields satisfying the properties of the definition of complex numbers, then call $\mathbb{C} and \mathbb{C}'$ the corresponding complex number fields.
Then $\mathbb{C} \cong \mathbb{C}'$.

**Proof:**
Call $i \in \mathbb{C}$ and $i' \in \mathbb{C}'$the square roots $-1$.
Then:

$$f : \mathbb{C} \to \mathbb{C}' \text{ defined as } f(\alpha + i\beta) = \alpha + i'\beta \text{ is an isomorphism}$$

$$z = \alpha + i\beta, z' = \alpha' + i'\beta', w = \gamma + i\delta, w' = \gamma' + i'\delta'$$

**Lemma 1.2.2**

Consider the vetor space $(\mathbb{R}, +)$.Then any multiplication '$\times$' defined in this vector space which transforms it into a field $(\mathbb{R}, +, \times)$ ,is such that $(\mathbb{R}, +, \times) \cong \mathbb{C}$.

**Proof:**
Consider the elements

$$(1, 0) \text{ and } j = (0, 1) \in \mathbb{R}^2$$

Then any elemnt of $\mathbb{R}^2$ can be written as$\alpha + j\beta, \alpha, \beta \in \mathbb{R}$
Then

$$j^2 = a + jb, a, b \in \mathbb{R}$$

Then,

$$j^2 - jb = a, j^2 - 2\frac{jb}{2} = a \to j^2 - 2\frac{jb}{2} + \frac{b^2}{4} = a + \frac{b^2}{4}$$

$$(j + \frac{b}{2})^2 = a + \frac{b^2}{4} \tag{1.1}$$

**Define:**$i = \frac{j - \frac{b}{2}}{\sqrt{a + \frac{b^2}{4}}}$
**Claim:**

$$a + \frac{b^2}{4} < 0$$

If:

$$a + \frac{b^2}{4} \geq 0$$

Then (1.1) can be written as

$$\to ((j - \frac{b}{2}) + \sqrt{a + \frac{b^2}{4}})((j - \frac{b}{2}) - \sqrt{a + \frac{b^2}{4}}) = 0$$

either

$$j - \frac{b}{2} = \sqrt{a + \frac{b^2}{4}}, j - \frac{b}{2} = -\sqrt{a + \frac{b^2}{4}}$$

## §1.3 Complex numbers

The field of rationals $\mathcal{Q}$ is the fractional field of the integral domian $\mathcal{Z}$.
With the metrice

$$d(x,y) = |x-y|$$
$$|x| = max\{x, -x\}, \forall x, y \in \mathcal{R}$$

The fielf $\mathcal{Q}$ is not a complete ,metrice space.
Its completion $\mathbb{R}$, nevertheless is not algebraically closed.
For instance, $f(x) = x^2 + 1$, doesn't split in $\mathbb{R}$.
So we define the spliting field for $f(x)$ over $\mathbb{R}$
i.e. the smallest field extension of $\mathbb{R}$ such that $f(x)$ has roots.

Consider the principle ideal $< f(x) >$ in $\mathbb{R}[x]$
since $f(x)$ is irreducible, then $< f(x) >$ si a maximal ideal.
Then the quotient ring $\mathbb{C} = \frac{\mathbb{R}[x]}{<f(x)>}$ is a field!
which is the spilting field of $f(x)$
The field $\mathbb{C}$ has no further spilting extensions! All polynomials in $\mathbb{C}[x]$ split in $\mathbb{C}$.
So as soon as we forced one polynomial. $f(x) = x^2 + 1$ to have roots, we in fact imposed that all polynomials have roots.

**Topology:**
The Topology of $\mathbb{C}$ is isomorphic to $\mathbb{R}^2$.
Thus $\mathbb{C}$ is locally compact, but not compact space.
The Alexandroff compactification says that we can add a point at infinity to $\mathbb{C}$ so that $\mathbb{C} \cup \{\infty\}$ compact.

**Definition 1.3.1: Alexandroff compactification**

- A set $U \supseteq \mathbb{C} \cup \{\infty\}$ is an open neighborhood of $\{\infty\}$ if $U = \mathbb{C} \cup \{\infty\}\ K$ When $K \supseteq \mathbb{C}$ .

$\mathbb{C} \cup \{\infty\}$ is called the Riemann sphere.

**Definition 1.3.2: Holomorphic Functions**

A function $f : \Omega \to \mathbb{C}$ is holomorphic where $\Omega \subseteq \mathbb{C}$ is holomorphic iff:

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = f'(z_0) \text{ exists}, \forall z_0 \in \Omega$$
$$f(z) = f(x,y) = u(x,y) + iv(x,y), wherez = x + iy \in \Omega$$
$$u(z) = Re(f(z)), v(z) = Im(f(z))$$

$f : \Omega \to \mathbb{C}$ being holomorphic of course implies that $u, v : \Omega \to \mathbb{R}^2$ differentiable.
If we take $f(z_0) = z_0 = 0$ and we write

$$\frac{\partial}{\partial z} = \frac{1}{2}(\frac{\partial}{\partial x} - i\frac{\partial}{\partial y}), \frac{\partial}{\partial \bar{z}} = \frac{1}{2}(\frac{\partial}{\partial x} + i\frac{\partial}{\partial y})$$

Then,
$$\frac{f(x)-f(z_0)}{z-z_0} = \frac{f(z)}{z} = (\frac{\partial f}{\partial z})_{(z_0=0)} + \frac{\bar{z}}{z}(\frac{\partial f}{\partial \bar{z}})_{(z_0=0)} + o(z)$$

For real $z \to 0$ , we have $\frac{\bar{z}}{z} = 1$ , and for purely imaginary $z \to 0$ , we have $\frac{\bar{z}}{z} = -1$.
we have to restrict $\frac{\partial f}{\partial z} = 0$
**Cauchy-Riemann equations:**

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \end{cases}$$

$$f'(z) = \frac{\partial f}{\partial z} = (\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x})(x,y), \text{where } z = x+iy \in \Omega$$

Nevertheless, $f$ being holomorphic is not equivalent to say $f$ satisfies the Cauchy-Riemann equations.
Indeed we take $f(z) = \frac{z^5}{|z|^4}$ on $\mathbb{C}\backslash\{0\}$ and $f(0) = 0$.
We can observe that $f(z)$ satisfies the Cauchy-Riemann equations on $\mathbb{C}$ but $f$ is not even coutinuous at $z = 0$.

**Theorem 1.3.1: Looman Menchoff Theorrem**

A continuous function $f : \Omega \cup \mathbb{C} \to \mathbb{C}$ is holomorphic iff $f$ satisfies the Cauchy-Riemann equations.

$$\begin{aligned} &\forall f, g \in H(\Omega) \\ &\qquad (fg) = f'g + fg' \\ &\qquad (\frac{f}{g}) = \frac{f'g - fg'}{g^2} \\ &(g \circ f)'(z) = g'(f(z)) \cdot f'(z) \end{aligned}$$

# §1.4 Lecture 3 (02-10) – Algebraic Structure of Complex numbers

## §1.4.1 Algebraic Structure of Complex numbers

**Lemma 1.4.1**

Consider $(\mathbb{R}^2, +)$ as a vector space. Then any definition of multiplication on $\mathbb{R}^2$ compatible with the vector space structure, which transforms $\mathbb{R}^2$ into a field, defines a field isomorphism with $\mathbb{C}$.

**Comments:**
The assumption that the multiplication is compatible with the vector space structure means that the scalar multiplication coincides with the field multiplication in the following space.
$$(\mathbb{R}^2, +, \times)$$

and $1 \in \mathbb{R}$ is the multiplicative identity
then

$$au = (a1) \times u, a \in \mathbb{R}$$

Let $L = \{a1 : a \in \mathbb{R}\}$
Then $L$ is isomorphic with $\mathbb{R}$
The addition in $\mathbb{R}^2$ as a vector space coincides with the addition in $\mathbb{R}^2$ as a field.
**Proof of the Lemma:**
Essentially we have to find a $i \in \mathbb{R}^2$ such that $i^2 = -1$
Since $\mathbb{R}^2$ is a two-dimentional vector space , if we pick any $j \in \mathbb{R}^2 \backslash L$
We know that $1, j, j^2$ must be linearly dependent.
Then thus must exist $a, b \in \mathbb{R}$ such that $aj^2 + bj + c1 = 0$
Then,

$$\begin{aligned} a(j^2) + 2\frac{b}{2a}j + c &= 0 \\ a(j + \frac{b}{2a})^2 + c - a(\frac{b}{2a})^2 &= 0 \\ (j + \frac{b}{2a})^2 &= \frac{1}{a}(-c + a(\frac{b}{2a})^2) \end{aligned}$$

Necesaarily, $-c + a(\frac{b}{2a})^2 < 0$
Define

$$i = \frac{\sqrt{a}(j + \frac{b}{2a})}{\sqrt{-a(\frac{b}{2a})^2 + c}}$$

### §1.4.2 Square roots

Every complex number has a square root.
Also we can write the 2 square roots in a quite explicit way(related to fundamental theorem of algebra).

**Lemma 1.4.2**

$$\sqrt{\alpha + i\beta} = \pm(\sqrt{\frac{\alpha + \sqrt{\alpha^2 + \beta^2}}{2}} + \frac{\beta}{|\beta|}\sqrt{\frac{-\alpha + \sqrt{\alpha^2 + \beta^2}}{2}})$$

**Proof:**
(Notation for complex numbers: $\alpha + i\beta, x + iy, \cdots$)
We need to find $z = x + iy$ such that $z^2 = (x + iy)^2 \alpha + i\beta$
So $x^2 - y^2 = \alpha, 2xy = \beta$
Now,

$$(x^2 + y^2)^2 = (x^2 - y^2)^2 + 4x^2y^2 = \alpha^2 + \beta^2$$

So,

$$
\begin{aligned}
x^2 + y^2 &= \sqrt{\alpha^2 + \beta^2} \\
x^2 - y^2 &= \alpha \\
\Rightarrow x^2 &= \frac{\alpha + \sqrt{\alpha^2 + \beta^2}}{2} \\
y^2 &= \frac{-\alpha + \sqrt{\alpha^2 + \beta^2}}{2} \\
x &= \pm\sqrt{\frac{\alpha + \sqrt{\alpha^2 + \beta^2}}{2}} \\
y &= \pm\sqrt{\frac{-\alpha + \sqrt{\alpha^2 + \beta^2}}{2}} \\
\sqrt{\alpha + i\beta} &= \pm(\sqrt{\frac{\alpha + \sqrt{\alpha^2 + \beta^2}}{2}} + \frac{\beta}{|\beta|}\sqrt{\frac{-\alpha + \sqrt{\alpha^2 + \beta^2}}{2}})
\end{aligned}
$$

**Definition 1.4.1: Algebraic Extension**

Let $F$ be a field and $E$ be a subfield of $F$.
We say that $x \in F$ is algebraic in the field $E$ if x satisfies the following equation:

$$a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 = 0 \text{ for } a_n, a_{n-1}, \cdots, a_1 \in E, an \neq 0$$

if every $x \in F$ is algebraic in $E$, then we say that $F$ is an algebraic extension of $E$.
We say that a field $F$ is algebraically closed if it does not have any strictly larger algebraic extension.
A field containing $F$ is called an algebraic closure if it is a closed algebraic extension of $F$ .

**Comments:**

- $\mathbb{R}$ is not an algebraic extension of $\mathbb{Q}$.
  If this was true, then every real number would be a root of a polynomial with rational coefficients:

  $$q_n x^n + q_{n-1}x^{n-1} + \cdots + q_1 x + q_0 = 0 \text{ (an algebraic number)}$$

  So the cardinality of the set of algebraic numbers is countable (trancsendental unmbers are real numbers which are not algebraic, e and $\pi$ for example ).

- $Q(\sqrt{2}) = \{q + \sqrt{2}q', q, q' \in Q\}$ is algebraic over $Q$.

- $\mathbb{R}$ is not algebraically closed.

- The fundamental theorem of algebra states that any nonconstant polynomial with complex coefficients has at least one complex root.
  Actually this is equivalent to saying that all the roots of such a polynomial are complex.

- The field of complex numbers is algebraically closed.

## §1.5 Lecture 4 (02-12) – Conjugation and absolute value

**Definition 1.5.1: Complex Conjugation**

Given a complex number $z = x + iy \in \mathbb{C}$, we define its complex conjugation $\bar{z} = x - iy$ This transformation is called complex conjugation.

**Comments:**

- $\bar{\bar{z}} = z$
- We define the real part of $z = x + iy$ as $Re(z) = x = \frac{z+\bar{z}}{2}$ and the imaginary part of z as $Im(z) = y = \frac{z-\bar{z}}{2i}$
- $\overline{z+w} = \bar{z} + \bar{w}, \overline{zw} = \bar{z}\bar{w}$: complex conjugation defines an isomorphism $\mathbb{C} \to \mathbb{C}$
- $\overline{(\frac{z}{w})} = \frac{\bar{z}}{\bar{w}}$
- if R is any rational function, then $\overline{R(z_1, z_2, \cdots, z_n)} = R(\bar{z_z}, \bar{z_2}, \cdots, \bar{z_n})$
  In particular,

$$\begin{aligned} a_n z^n + \cdots + a_1 z + a_0 = 0 \\ \Rightarrow \bar{a_n} \bar{z}^n + \cdots + \bar{a_1} \bar{z} + \bar{a_0} = 0 \end{aligned}$$

**Definition 1.5.2: absolute value**

Given a complex number $z \in \mathbb{C}$, we define its absolute value, denoted by

$$|z| = \sqrt{z\bar{z}}$$

**Comments:**

- If z is real, then $\bar{z} = z$
- we say that z is purely imaginary if $Re(z) = 0$, iff $\bar{z} = -z$
- Then if z is real, |z| coincides with the traditional definition of absolute value on $\mathbb{R}$
- $z\bar{z} = (x + iy)(x - iy) = x^2 - (iy)^2 = x^2 + y^2 \geq 0$
- Vector space (V,+,scalar multiplication) over a field $\mathbb{F}(Q, R, C)$
- $|zw| = |z||w|, |\bar{z}| = |z|$
- $|\frac{z}{w}| = \frac{|z|}{|w|}, w \neq 0$
- $|z_1 z_2 \cdots z_n| = |z_1||z_2| \cdots |z_n|$
- $|z + w|^2 = (z + w)(\bar{z} + \bar{w}) = |z|^2 + |w|^2 + z\bar{w} + \bar{z}w = |z|^2 + |w|^2 + 2Re(z\bar{w})$
- $|z + w|^2 + |z - w|^2 = 2|z|^2 + 2|w|^2$

**Lemma 1.5.1: Triangle inequality**

If $z, w \in \mathbb{C}$ then, $|z+w| \leq |z| + |w|$

**Proof:**
Note that if $a = \alpha + i\beta$ then

$$-|a| \leq Re(a) = \alpha \leq |a| = \sqrt{\alpha^2 + \beta^2}$$
$$\Rightarrow |z+w|^2 = |z|^2 + |w|^2 + 2Re(z\bar{w}) \leq |z|^2 + |w|^2 + 2|z||w| = (|z| + |w|)^2$$

**Comments:**
Assume that z and w are such that $|z+w| = |z| + |w|$

$$\Rightarrow |z||w| = Re(z\bar{w}) \Leftrightarrow z\bar{w} \geq 0$$

We proved that absolute value is a norm (which is a good notion of distance):

- $|z| = 0 \Leftrightarrow z = 0$ and $|z| \geq 0, \forall z \in \mathbb{C}$
- $|az| = |a||z|, \forall a, z \in \mathbb{C}$
- $|z+w| \leq |z| + |w|$
- $(R^d, +, \underbrace{|| \; ||}_{\text{Euclidean norm}}), ||(x_1, \cdots, x_d)|| = \sqrt{{x_1}^2 + \cdots + {x_d}^2}$

**Lemma 1.5.2: Cauchy inequality**

If $z_1, \cdots, z_n, w_1, \cdots, w_n \in \mathbb{C}$ ,

$$|\sum_{i=1}^{n} z_i w_i| \leq \sqrt{\sum_{i=1}^{n} |z_i|^2} \sqrt{\sum_{i=1}^{n} |w_i|^2}$$

**Proof:**
Let $\lambda \in \mathbb{R}$ and consider

$$\sum_{i=1}^{n} |z_i - \lambda w_i|^2 = \sum_{i=1}^{n} |z_i|^2 - 2Re\bar{\lambda}(\sum_{i=1}^{n} z_i \bar{w_i}) + |\lambda|^2 \sum_{i=1}^{n} |w_i|^2$$

Choose

$$\lambda = \frac{\sum_{i=1}^{n} z_i \bar{w_i}}{\sum_{i=1}^{n} |w_i|^2}$$

Then,

$$0 \leq \sum_{i=1}^{n} |z_i|^2 - \frac{|\sum_{i=1}^{n} z_i \bar{w_i}|^2}{\sum_{i=1}^{n} |w_i|^2}$$

Chapter 2

# Honors Theory of Probability

## §2.1 Lecture 1 (02-03)Introduction to Probability

Office Hour: Wed 12:30-1:30pm,Fri 3:30-4:30pm W910

- Homework: weekly
- Grades: 5% participation, 15% homework, 40% midterm, final group project(Max 4 people a group,presetation 20%, Final report 20% 10 pages)

### §2.1.1 Intro

Why do we need modern theory of probability?

**Example 1**

- Coin flip 7 times, what is P[first outcome and fifth outcome are Head]?
sol:

$$\Omega = (x1, x2, x3, x4, x5, x6, x7)$$

$$A = (x1, x2, x3, x4, x5, x6, x7), x1 = x5 = H$$

$$P(A) = \frac{|\Omega|}{|A|} = \frac{1}{4}$$

- Stock Price (mathematically,geometric Browmion motion, STochastic process, such that $t \to S_t$ is continuous but nowehere differentiable)
What is P[$S_T \geq 100$]?

$$\Omega = C[0, T] = \{\text{continuous function}, f[0, T] \to R\}$$

$$A = \{f \in C[0, T] | f(t) \geq 100\}$$

requires measure theory that defines P[$S_T \geq 100$]
In modern Prob, Probability Space:$(\Omega, F, P)$, $\Omega$ is sample space, $F$ is $\sigma$-algebra(meaningful subset of $\sigma$), $P$ is probability measure($P : F \to [0, 1]$).

Topics covered:

- Probability space, $\sigma$-algebra, measure ,Conditional Probability and Independence
- Random variables (measurable functions),distribution
- Expectation (Lebesgue integral),Conditional distribution and expectation, functionos of random variables, Radom-Nikodym Derivative
- Random walks

- generating functions, characteristic functions
- Branching processes
- Convergence of random variables, Law of large numbers, Morte-Carlo Method
- Central Limit Theorem
- Time permitting: Large deviations, Markov Chains

### §2.1.2 Probability Space

$(\Omega, F, P)$

**Example 2: Coinflip and Stock Price**

- Coin flip infinite times,

$$\Omega = \{(x1, x2, \cdots), x_i = H, T\}$$

Let $A$ be the event of $10^6$ consecutive Tails,

$$A = \bigcup_{i=1}^{+\infty} \{x_i = x_{i+1} = \cdots = x_{i+10^6-1} = 0, (x_i \in \Omega)\}$$

$P[A] = 1$

- Discrete Stock Price model, t=0,1,2,$\cdots$,T
T is maturing time, time step $\Delta t \ll T$

$$N = \frac{T}{\Delta t}$$

$$\text{price go} \begin{cases} \nearrow \text{by factor} : e^{\sigma\sqrt{\Delta t}} \\ \searrow \text{by factor} : e^{-\sigma\sqrt{\Delta t}} \end{cases}$$

$$\Omega = \{(x_1, x_2, \cdots, x_N), x_i = 0 \text{ or } 1\}$$

Stock price at time t: $\forall \omega \in \Omega$

$$S_N(\omega) = S_0 e^{\sum_{i=1}^N x_i \sigma\sqrt{\Delta t} e^{(N - \sum_{i=1}^N x_i)(-\sigma\sqrt{\Delta t})}}$$

$$S : \Omega \to R(\text{Random Variable})$$

Event: return at T is positive but not more than 10%:

$$\{\omega \in \Omega : S_N(\omega) \in (S_0, 1.1S_0]\}$$

**Example 3: Gambling**

- Gambling:

- start with {0,1,2,$\cdots$} each time bet an interger amount
- if amount of money =0, stays at 0
  wealth process: $\Omega - \{0,1,2,\cdots\} \times \{0,1,2,\cdots\} \times \cdots$
  wealth after time n: (random variable) $X_n : \Omega \to N, (x_1, x_2, ...x_n) \to x_n$
  $(X_n)$is a Markov Chain (future only depends on present state, but not the past)

$$\begin{aligned} &\text{Event: } \{\text{State j is reached from state i}\} \\ &= \{\omega \in \Omega : \exists n \in N, X_0(\omega) = i, X_n(\omega) = j\} \\ &= \bigcup_{m=1}^{+\infty} \{\omega \in \Omega : X_o(\omega) = i, X_n(\omega) = j\} \end{aligned}$$

## §2.2 Lecture 2 (02-05) – Algebra

### §2.2.1 algebra and $\sigma$-algebra

in practice, want $F$ to be closed under $\bigcap, \bigcup, ^c$

**Definition 2.2.1**

Let $\mathcal{A}$ be a collection of subsets of $\Omega$ , $\mathcal{A}$ is an algebra iff:

- $\Omega \in \mathcal{A}$
- if A,B$\in \mathcal{A}$ then A$\cap$B $\in \mathcal{A}$
- if A$\in \mathcal{A}$ then A$^c \in \mathcal{A}$

**Remark 1**

if A,B$\in \mathcal{A} \to$ A $\cap$ B$\in \mathcal{A}$, because A$\cap$B = (A$^c \cup$B$^c)^c$

**Fact:**

① $P(\Omega)$(powerset)={A:A $\subset \Omega$} is an algebra

② smallest algebra/trivial algebra: $\{\emptyset, \Omega\}$

③ Let $\mathcal{A}_1, \mathcal{A}_2$ be two algebras of $\Omega$

$$\mathcal{A}_1 \cap \mathcal{A}_2 = \{B \in \Omega : B \in \mathcal{A}_1 \text{ and } B \in \mathcal{A}_2\} \text{ is an algebra}$$

$$\text{if}(\mathcal{A}_j)_{j\in J} \text{is a family of algebras, then } \bigcap_{j\in J} \mathcal{A}_j \text{ is an algebra}$$

④ Let $\mathcal{E}$ be any collection of subsets of $\Omega$

$$a(\mathcal{E}) = \bigcap_{\substack{\mathcal{A}\supseteq\mathcal{E} \\ \mathcal{A} \text{ is an algebra}}} \mathcal{A} \text{ is an algebra}$$

$a(\mathcal{E})$ is an algebra by ③
in fact, $a(\mathcal{E})$ is the smallest algebra containing $\mathcal{E}$
It is called the algebra generated by $\mathcal{E}$

⑤ Let $A \subseteq \Omega$ ,$\mathcal{E} = \{A\}$
Then

$$a(\mathcal{E}) = \{\underbrace{A, A^c, \Omega, \emptyset}_{f}\}$$

**Proof:**

- $a(\mathcal{E}) \subseteq f$
  notice that $f$ is an algebra since $f \supseteq \mathcal{E}$, therefore $f \supseteq a(\mathcal{E})$ because $a(\mathcal{E})$ is the smallest algebra containing $\mathcal{E}$.
- $f \subseteq a(\mathcal{E})$:

$$A \in a(\mathcal{E}), A^c \in a(\mathcal{E})$$

  beacuse $a(\mathcal{E})$ is an algebra, $\emptyset, \Omega \in a(\mathcal{E})$

⑥ $\pi = \{A_1, A_2, \cdots, A_n\}$, $\Omega = \bigcup_{i=1}^n A_i, A_i \cap A_j = \emptyset$ Then:

$$a(\pi) = \{\bigcup_{i \in I} A_i, for I \subset 1, 2, \cdots, n\} = \text{ finite disjoint union of } (A_i)_{i=1}^n$$

⑦ Let $\mathcal{A}$ be an algebra of $\mathbb{R}$
Let $X : \Omega \to$ be a function
Then

$$\underbrace{\{X^{-1}(A), A \in \mathcal{A}\}}_{\{\omega \in \Omega : X(\omega) \in A, \text{ for some } A \in \mathcal{A}\}} \qquad \text{is an algebra of } \Omega$$

**Hint:**

$$X^{-1}(A \cup B) = (X^{-1}(A)) \cup (X^{-1}(B))$$

⑧ $\Omega = \mathbb{R}, \mathcal{E} = \{\text{left open right closed intervals}\} = \begin{cases} (a, b], & -\infty \leq a < b < +\infty \\ (a, +\infty), a \in \mathbb{R} \end{cases}$

Then:

$$\begin{aligned} a(\mathcal{E}) = & \text{ "finite disjioint union of elements in } \mathcal{E}\text{"} \\ = & \underbrace{\{I_1 \cup \cdots \cup I_k, I_j \in \mathcal{E}, I_j \cup I_j = \emptyset(f)\}}_{f} \end{aligned}$$

**Hint:**

- $f \subseteq a(\mathcal{E})$ strightforward
- $a(\mathcal{E}) \subseteq f$
  - check $f$ is an algebra
  - since $\mathcal{E} \subset f$ , $a(\mathcal{E})$ is the smallest algebra containing $\mathcal{E}$, $a(\mathcal{E}) \subseteq f$
  - $(a, b]^c = \underbrace{(-\infty, a] \cup (b, +\infty)}_{\in f}$

**Lemma 2.2.1**

In Probability, if $(A_k)_{k=1}^n$ are disjoint events, we have $P(\bigcup_{k=1}^{+\infty} A_k) = \sum_{k=1}^{+\infty} P(A_k)$

**Definition 2.2.2:** $\sigma - algebra$

A $\sigma - algebra$ ($\sigma - field$) $\mathcal{A}$ is an collection of subsets of $\Omega$ such that

- $\Omega \in \mathcal{A}$
- if $A_1, A_2, \cdots \in \mathcal{A}$ then $\bigcup_{i=1}^{+\infty} A_i \in \mathcal{A}$
- if $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$

**Remark 2**

if $A_1, A_2, \cdots \in \mathcal{A}$ then $\bigcap_{i=1}^{+\infty} A_i \in \mathcal{A}$
$\sigma$ -algebra represents the collection of information determined by partial derivatives

**Example 1**

- coinflips infinity many times

$$\Omega = \{(x_1, x_2, \cdots)x_i = 0, 1\} = (0, 1)^{\infty}$$

After observing first outcome, $f_1 = \{\emptyset, \Omega, A_0, A_1\}$
Where $A_0 = \{(0, x_2, x_3, \cdots), x_i = 0, 1\}, A_1 = \{(1, x_2, x_3, \cdots), x_i = 0, 1\}$

After observing second outcome, $f_2 = \{\emptyset, \Omega, A_0, A_1, A_00, A_01, A_10, A_11\}$
Where $A_{00} = \{(0, 0, x_3, x_4, \cdots), x_i = 0, 1\}, A_{01} = \{(0, 1, x_3, x_4, \cdots), x_i = 0, 1\} \cdots$

After observing first n outcomes,

$$f_n = \{\emptyset, \Omega, (A_j)_{j \in (\sigma 1)^n} \text{and finite disjioint unions}\}$$

and

$$f_1 \subseteq f_2 \subseteq \cdots \subseteq f_n \subseteq \cdots$$

**Proposition 2.2.0**

① intersection of $\sigma$-algebras is a $\sigma$-algebra

② Let $\mathcal{E}$ be a collecion of subsets of $\Omega$

$$\sigma(\mathcal{E}) = \bigcap_{\substack{f \supseteq \mathcal{E} \\ f \text{ is an algebra}}} f \text{ is a smallest } \sigma\text{-algebra that contains } \mathcal{E}$$

③ For any collection $\mathcal{E}$, we have $a(\mathcal{E}) \subseteq \sigma(\mathcal{E})$

④ For any collection $\mathcal{E}$, we have $\sigma(a(\mathcal{E})) = \sigma(\mathcal{E})$
**Hint:**

$$a(\mathcal{E}) \subseteq \sigma(\mathcal{E}) \to \sigma(a(\mathcal{E})) \subseteq \sigma(\mathcal{E})$$
$$\mathcal{E} \subseteq \sigma(a(\mathcal{E})) \to \sigma(\mathcal{E}) \subseteq \sigma(a(\mathcal{E}))$$

# §2.3 Recitation 1 (02-07) – Problem Solving

## §2.3.1 Basic Set Theory

**Definition 2.3.1:** $\cup, \cap, ^c$

**De Morgan's Law:**

$$(\bigcup_{j\in J} A_j)^c = \bigcap_{j\in J} A_j^c$$
$$(\bigcap_{j\in J} A_j)^c = \bigcup_{j\in J} A_j^c$$

**Definition 2.3.2:** $\backslash$

$A\backslash B = A \cap B^c$
Then $A\backslash B = A \cap B^c = A\backslash(A \cap B) = B^c\backslash A^c$

**Remark 1**

$$\bigcap_{n\geq 1} A_n = A_1\backslash(\bigcup_{j\geq 2}(A_1 - A_j)) \text{ (ex)}$$

## §2.3.2 Limits of Sets

**Definition 2.3.3: Limit Sets**

Let $(A_n)_{n\geq 1}$ be a sequence of sets, then

$$B_k = \bigcup_{n\geq k} A_n, C_k = \bigcap_{n\geq k} A_n$$

Then $B_k$ is increasing, $C_k$ is decreasing
Define:

$$\limsup_{n\to+\infty} A_n = \lim_{k\to+\infty} B_k = \bigcap_{k\geq 1}\bigcup_{n\geq k} A_n$$
$$\liminf_{n\to+\infty} A_n = \lim_{k\to+\infty} C_k = \bigcup_{k\geq 1}\bigcap_{n\geq k} A_n$$

**Definition 2.3.4: liminf, limsup of sequence**

$$\limsup an = \inf_{k\geq 1}\sup_{n\geq k} a_n, \liminf an = \sup_{k\geq 1}\inf_{n\geq k} a_n$$

When $\limsup A_n = \liminf A_n$, Then we say $\lim_{n\to+\infty} A_n$ exist and $\lim_{n\to+\infty} A_n = \limsup A_n = \liminf A_n$

In probability,

$$\begin{aligned}
\limsup A_n &= \{A_n \text{ occurs infinitely often}\}\\
&= \{An.i.o\}\\
x \in \limsup A_n &\Leftrightarrow \forall k \in \mathbb{N}, \exists n \geq ksuchthatx \in A_n\\
&\Leftrightarrow A_n.i.o\\
\liminf A_n &= \{A_n \text{ occurs eventually}\}\\
&\Leftrightarrow \exists k \in \mathbb{N}, \forall n \geq k, x \in A_n
\end{aligned}$$

**Remark 2**

1. if $A_n$ increases, then $\limsup A_n = \lim_{n\to\infty} A_n = \bigcup_{n\geq 1} A_n$
   if $A_n$ decreases, then $\liminf A_n = \lim_{n\to\infty} A_n = \bigcap_{n\geq 1} A_n$
2. $$(\limsup A_n)^c = \liminf A_n, (\liminf A_n)^c = \limsup A_n$$

## §2.3.3 Exercise

1. $A_k = \begin{cases} E, \text{if k is odd} \\ F, \text{if k is even} \end{cases}$
   Then $\limsup A_n = E \cup F, \liminf A_n = E \cap F$
2. Let $f_n : \mathbb{R} \to \mathbb{R}$ , Let A=$\{x \in \mathbb{R} : \lim_{n\to\infty} f_n(x) = f(x)\}$ Then,
   $$\begin{aligned}
   A^c &= \bigcup_{k=1}^{+\infty}\bigcap_{m=1}^{+\infty}\bigcup_{n=m}^{+\infty}\{x : |f_n(x) - f(x)| \geq \frac{1}{k}\}\\
   &= \bigcup_{k=1}^{+\infty}[\limsup_n\{x : |f_n(x) - f(x)| \geq \frac{1}{k}\}]
   \end{aligned}$$
3. Suppose that $\lim_{n\to\infty} f_n(x) = f(x), \forall x \in \mathbb{R}$
   Then
   $$\begin{aligned}
   \{x : f(x) \leq t\} &= \bigcap_{k=1}^{+\infty}\bigcup_{m=1}^{+\infty}\bigcup_{n=m}^{+\infty}\{x \in \mathbb{R} : f_n(x) < t + \frac{1}{k}\}\\
   &= \bigcap_{k=1}^{+\infty}[\liminf_n\{x : f_n(x) < t + \frac{1}{k}\}]
   \end{aligned}$$

**Proof:**

1. ex2: Want

$$A = \bigcup_{k\geq 1}^{+\infty} \bigcap_{\geq =1}^{+\infty} \bigcup_{n\geq m}^{+\infty} \{x : |f_n(x) - f(x)| \geq \frac{1}{k}\}$$
$$f_n(x) \to f(x) \text{ iff } \forall \epsilon > 0, \exists N \geq N, |f_n(x) - f(x)| < \epsilon$$
$$\{x : f_n(x) \to f(x)\} = \bigcap_{\epsilon>0} \bigcup_{N} \bigcap_{n\geq N} \{x : |f_n(x) - f(x)| < \epsilon\},$$
$$\text{USE that } \{|f_n - f| < \epsilon\} \text{ is monotone increasing in } \epsilon$$

**Review on mapping:**

$$f : X \to Y, f^{-1} : Y \to X$$

**Basic properties:**

- $f(\bigcup_{i\in I}) = \bigcup_{i\in I} f(A_i)$
- $f(\bigcap_{i\in I}) = \bigcap_{i\in I} f(A_i)$

For $(B_i)$ subset of Y:

- if $B_1 \subset B_2, f^{-1}(B_1) \subset f_-1(B_2)$
- $f^{-1}(\bigcup_{i\in I} B_i) = \bigcup_{i\in I} f^{-1}(B_i)$
- $f^{-1}(B^c) = (f^{-1}(B))$

**Definition 2.3.5: Indicator Mapping**

$$1_A : X \to \{0, 1\}$$
$$1_A(x) = \begin{cases} 1, x \in A \\ 0, x \notin A \end{cases}$$

**Exercise:**

1. $1_{limsupA_n} = limsup 1_{A_n}$
2. $1_{liminfA_n} = liminf 1_{A_n}$

**Proof:**
if

$$\begin{aligned} 1_{limsupA_n}(x) = 1 &\Leftrightarrow x \in limsupA_n \\ &\Leftrightarrow \forall k \in \mathbb{N}, \exists n \geq k, x \in A_n \\ &\Leftrightarrow \forall k \in \mathbb{N}, \exists n \geq k, 1_{A_n}(x) = 1 \\ &\Leftrightarrow limsup 1_{A_n}(x) \geq 1 (\text{Definition of limsup}) \\ &\Leftrightarrow 1_{limsupA_n}(x) = 1 \end{aligned}$$

**Exercise:** Let $\mathcal{A}_1, \mathcal{A}_2$ be algebras of $\Omega$

1. show that $\underbrace{\mathcal{A}_1 \cap \mathcal{A}_2}_{B\subset\Omega : B\in\mathcal{A}_1, B\in\mathcal{A}_2}$ is an algebra

2. show that $\underbrace{\mathcal{A}_1 \cup \mathcal{A}_2}_{B\subset\Omega: B\in\mathcal{A}_1 \text{ or } B\in\mathcal{A}_2}$ is an algebra iff $\mathcal{A}_1 \subseteq \mathcal{A}_2$ or $\mathcal{A}_2 \subseteq \mathcal{A}_1$

**Proof:** Suppose by contradiction that

$$\exists A_1 \in \mathcal{A}_1 \text{ but } A_1 \notin \mathcal{A}_2 \text{ and } A_2 \in \mathcal{A}_2 \text{ but } A_2 \notin \mathcal{A}_1 \text{ and } \mathcal{A}_1 \cup \mathcal{A}_2 \text{ is an algebra}$$

Therefore:

$$A_1 \cup A_2 \in \mathcal{A}_1 \cup \mathcal{A}_2, A_1 \backslash A_2 = A_1 \cap \underbrace{A_2^c}_{\mathcal{A}_1\cup\mathcal{A}_2} \in \mathcal{A}_1 \cup \mathcal{A}_2$$

$$A_2 \backslash A_1 = A_2 \cap A_1^c \in \mathcal{A}_1 \cup \mathcal{A}_2$$

$\Rightarrow$ at least two of

$(A_1\backslash A_2) \cup (A_2 \backslash A_1), A_1\backslash A_2, A_2\backslash A_1$ are in $\mathcal{A}_1$ or $\mathcal{A}_2$

**Assume in $\mathcal{A}_1$**

$\Rightarrow$ (by $\mathcal{A}_1$ is an algebra, all three sets are in $\mathcal{A}_1$)

$\Rightarrow A_2 = (\underbrace{A_1 \cup A_2}_{\in\mathcal{A}_1})\backslash(\underbrace{A_1\backslash A_2}_{\mathcal{A}_1}) \in \mathcal{A}_1$

**Contradiction!**

## §2.4 Lecture 3 (02-10)

**Recall $\sigma$-algebra:**

**Example 1**

1. We know $\Omega = \mathbb{R}$,

$\mathcal{E} = \{\text{left open right closed intervals}\} = \begin{cases} (a, b], & -\infty \leq a < b < +\infty \\ (a, +\infty), & a \in \mathbb{R} \end{cases}$

Then we know:

$$a(\mathcal{E}) = \{ \text{"finite disjioint union of elements in } \mathcal{E}\text{"}\}$$

What is $\sigma(a(\mathcal{E}))$?
$\sigma(\epsilon) =$ Borel Sets $\mathcal{B}(R)$
Any "reasonable" subset of $\mathbb{R}$ is in $\sigma(\epsilon)$

- $(a, b) \in \sigma(\epsilon) : (a, b) = \bigcup_{n\geq 1} \underbrace{(a, b - \frac{1}{n}]}_{\in\sigma(\epsilon)} \in \sigma(\epsilon)$
- any singleton $\{a\} \in \sigma(\epsilon)$, because $\{a\} = \bigcap_{n\geq 1} \underbrace{(a - \frac{1}{n}, a + \frac{1}{n})}_{\in\sigma(\epsilon)} \in \sigma(\epsilon)$
- any countable set is in $\sigma(\epsilon)$ (e.g. $\mathbb{Q}$)
- The set of trancsendental numbers is in $\sigma(\epsilon)$, because the set of algebraic numbers is countable

**Definition 2.4.1: measurable set**

A pair $(\Omega, F)$ , where $F$ is a $\sigma$-algebra of $\Omega$, is called a measurable space
Any set $A \in F$ is called a measurable set

## §2.4.1 Content and Measure

**Definition 2.4.2: Content**

Let $\mathcal{A}$ be an algebra of $\Omega$, A set function $\mu : \mathcal{A} \to [0, +\infty)$ is called a content iff:

- $\mu(\emptyset) = 0$
- if A,B$\in \mathcal{A}$ and $A \cap B = \emptyset$, then $\mu(A \cup B) = \mu(A) + \mu(B)$ (finite additivity)

**Lemma 2.4.1**

Let $\mu : \mathcal{A} \to [0, +\infty)$ be a content, $\forall A, B \in \mathcal{A}$ then:

① $\mu(A) + \mu(B) = \mu(A \cup B) + \mu(A \cap B)$

② if $A \subset B$, and $\mu(A) < +\infty$ ,then $\mu(B \backslash A) = \mu(B) - \mu(A)$

③ if $A \subset B$, then $\mu(A) \leq \mu(B)$

④ if $A_1, A_2, \cdots \in \mathcal{A}$ , then $\mu(\bigcup_{i=1}^n A_i) = \sum_{i=1}^n \mu(A_i)$

⑤ if $A_1, A_2, \cdots \in \mathcal{A}$ , $A_i \cap A_j = \emptyset$ and $\bigcup_{i=1}^{+\infty} \in A$ then $\mu(\bigcup_{i=1}^{+\infty} A_i) \geq \sum_{i=1}^{+\infty} \mu(A_i)$

**Proof:**
finite add:
1:

$$\begin{aligned} \mu(B) &= \mu(A \cup B) + \mu(B \backslash A) \\ \mu(A) + \mu(B \backslash A) &= \mu(A \cup B) \\ \Rightarrow \mu(A) + \mu(B) &= \mu(A \cup B) + \mu(A \cap B) \end{aligned}$$

4:

$$\begin{aligned} \text{Let } B_1 = A_1, B_2 = A_2 \backslash A_1, B_3 = A_3 \backslash (A_1 \cup A_2), \cdots \\ \text{Then } B_j \text{ are disjoint and } \bigcup_{i=1}^n B_i = \bigcup_{i=1}^n A_i \\ \text{By finite add for } B_j : \\ \mu(\bigcup_{i=1}^n A_i) = \mu(\bigcup_{i=1}^n B_i) = \sum_{i=1}^n (\mu(A_i) \backslash \bigcup_{i=1}^{j=1} A_j) \leq \sum_{i=1}^n \mu(A_i) \end{aligned}$$

5:

For $\forall n$,

$$\mu(\bigcup_{i=1}^{+\infty} A_i) \geq \mu(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mu(A_i) \text{ send n } \nearrow +\infty \text{ to conclude}$$

**Remark 1**

In general. $\mu(\bigcup_{i=1}^{+\infty} A_i) \neq \sum_{i=1}^{+\infty} \mu(A_i)$ although $\mu(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mu(A_i)$ require continuity of $\mu$

**Counterexample:**

- Let $\Omega = \mathbb{R}, \mathcal{A} = a(\epsilon), \forall A \in \mathcal{A}$,

$$\mu(A) = \lim_{L \to +\infty} \frac{\overbrace{|A \cup [0, L]|}^{\text{Length of interval}}}{L} \text{ "Density of A in } [0, +\infty)\text{"}$$

Then $\mu$ is a content.
Take $A_i = (i, i+1]$ , then $\mu((i, i+1]) = 0$, But $\mu(\bigcup_{i=0}^{+\infty}) = \mu((0, +\infty)) = 1$

- For $\mathbb{R}, a(\epsilon)$, Given $A \in a(\epsilon)$

$$\mathring{A} = \{x \in A : \exists r_x > 0, s.t. (x - rx, x + rx) \subset A\}$$
$$\partial A = \bar{A} \backslash \mathring{A}$$

$$\mu(A) = \begin{cases} 2, & \text{if } 0 \in \mathring{A} \\ 1, & \text{if } 0 \in \partial A \\ 0, & \text{else} \end{cases}$$

Then $\mu$ is a content
However, $A_i = (\frac{1}{i+1}, \frac{1}{i}], \mu(A_i) = 0,$ But $\mu(\bigcup_{i=1}^{+\infty}) = \mu((0, 1])$

**Example:**

- (Discrete Probability)

$$\Omega = \{\omega_1, \cdots, \omega_n\} \text{ finite set, } , F = P(\Omega)$$

Set $A_i = \{\omega_i\}, P(A_i) = P(\omega_i) = P_i$ such that $\sum_{i=1}^{n} P_i = 1$
Then $P : P(\Omega) \to [0, 1]$ defines a content on $P(\Omega)$ by extending the P using finite additivity:

$$\forall A \in P(\Omega), P(A) = \sum_{\omega \in A} P(\omega)$$

- $\Omega = \mathbb{R},$ algebra $\mathcal{A} = a(\epsilon) = \{$ finite disjoint umiom of elements in $\epsilon\}$
Define $m : a(\epsilon) \to [0, +\infty)$ , set $m([a, b])$ =b-a,
and extend by additivity:

$$m(I) = \sum_{i=1}^{n} m(I_j), \text{ if } I = I_1 \cup \cdots \cup I_n, I_j \cap I_i = \emptyset$$

$\Rightarrow$ m is a content on $(\mathbb{R}, a(\epsilon))$
m can be further extended to $(\mathbb{R}, \sigma(\epsilon))$, called Lebesgue Measure

**Definition 2.4.3: countably additive**

A content $\mu : \mathcal{A} \to [0, +\infty)$ is countably additive if:

$$\mu(\bigcup_{i=1}^{+\infty} A_i) = \sum_{i=1}^{+\infty} \mu(A_i) \text{ for every disjioint } A_1, A_2, \cdots \in \mathcal{A}$$

**Definition 2.4.4: measure**

Let $(\Omega, F)$ be an measurable space, then a content $\mu : F \to [0, +\infty)$ that is countably additive is called a measure

**Lemma 2.4.2**

$m : a(\epsilon) \to [0, +\infty)$ is countably additive

**Proof:**
Let $A_1, A_2, \cdots \in a(\epsilon), (A_k)$ disjioint , $A := \bigcup_{i=1}^{+\infty} A_i \in a(\epsilon)$
Want to show :

$$m(A) = \sum_{i=1}^{+\infty} m(A_i)$$

we can write, using $A, (A_k) \in a(\epsilon)$, $A = \bigcup_{j=1}^{n} I_j$, where $I_j \in \epsilon$ and $(I_j)$ disjioint
$A_j = \bigcup_{k=1}^{n_i} J_{ik}$, where $J_{ik} \in \epsilon$,$(J_{ik})$ disjioint
Then:

$$\begin{aligned}
m(A) &= \sum_{j=1}^{n} m(I_j) \\
&= \sum_{j=1}^{n} \sum_{j=1}^{+\infty} \sum_{k=1}^{n_i} m(I_j \cap J_{ik}) \\
&= \sum_{i=1}^{+\infty} m(\underbrace{\bigcup_{j=1}^{n} I_j}_{A} \cap (\underbrace{\bigcup_{k=1}^{n_i} J_{i_{jk}}}_{A_i})) \\
&= \sum_{i=1}^{+\infty} m(A_i)
\end{aligned}$$

## §2.5 Lecture 4 (02-12)

$$\Omega = \mathbb{R}, m : \epsilon \to [0, +\infty), \text{ such that } \begin{cases} m([a,b]) = b-a, \\ m((a,+\infty)) = +\infty \end{cases}$$

$$\text{extend m to } a(\epsilon) : \forall A \in a(\epsilon),$$

$$\text{if } A = \bigcup_{i=1}^{n} I_j, I_j \text{ disjioint}, \ m(A) = \sum_{j=1}^{n} m(I_j)$$

**Fact:**
if $I \in \epsilon$ s.t. $I = \bigcup_{i=1} +\infty I_i, (I_i)$ disjoint and $I_j \in \epsilon$
Then

$$\begin{aligned} m(I) &= |I| \\ &= \sum_{i=1}^{+\infty} |I_i| = \bigcup_{i=1}^{+\infty} m(I_i) \end{aligned}$$

**Lemma 2.5.1**

we want to prove m is a countably additive content on $(\mathbb{R}, a(\epsilon))$

$$\text{Let } A_j \in a(\epsilon), A = \bigcup_{j=1}^{\infty} \in a(\epsilon), (A_j) \text{ disjoint}$$

$$\exists (I_i) \text{ such that } I_i \in \epsilon, \ \text{disjoint } A = \bigcup_{i=1}^{n} I_i$$

$$\exists (J_{ij}) \text{ such that } J_{ij} \in \epsilon, \ \text{disjoint } A_j = \bigcup_{k=1}^{j} J_{ik}$$

$$\begin{aligned} m(A) \underbrace{=}_{\text{def}} \sum_{i=1}^{n} m(I_i) &= \sum_{i=1}^{n} m(\bigcup_{j,k} \underbrace{(I_i) \cap (J_{ik})}_{\in \epsilon}) \\ &\underbrace{=}_{\text{fact}} \sum_{i=1}^{n} \sum_{j=1}^{+\infty} \sum_{k=1}^{n_j} m(I_i \cap I_{jk}) \\ &\underbrace{=}_{\text{finite add}} \sum_{j=1}^{+\infty} m(A_j) \end{aligned}$$

**Theorem 2.5.1**

m extends to a measure on $(\mathbb{R}, \sigma(\epsilon) = \mathcal{B}(\mathbb{R}))$ It is the unique measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that

$$m([a,b]) = b - a$$

**Definition 2.5.1**

If $\mu$ is a measure on $(\Omega, F)$, then $(\Omega, F, \mu)$ is said to be a measure space
If $\mu(\Omega) = 1$, then $(\Omega, F, \mu)$ is said to be a probability space

**Lemma 2.5.2**

Let $(\Omega, F, \mu)$ be a measure space, then:

① stability: Let $A_1, A_2, \cdots \in F$, then $\mu(\bigcup_{i=1}^{+\infty} A_i) \leq \sum_{i=1}^{+\infty} \mu(A_i)$

② continuity from below: Let $A_1, A_2, \cdots \in F$, $A_1 \subseteq A_2 \subseteq \cdots$, then

$$\mu(\bigcup_{i=1}^{+\infty} A_i) = \lim_{i\to+\infty} \mu(A_i) = \mu(\lim_{i\to+\infty} A_i)$$

③ continuity from above: Let $A_1, A_2, \cdots \in F$, $A_1 \supseteq A_2 \supseteq \cdots$,
$(\mu(A_i) < +\infty$ we need this for the Counterexample $A_i = [i, +\infty)$ ) then

$$\mu(\bigcap_{i=1}^{+\infty} A_i) = \lim_{i\to+\infty} \mu(A_i) = \mu(\lim_{i\to+\infty} A_i)$$

**Proof:**

① Let $B_1 = A_1, B_2 = A_2 \backslash A_1, \cdots, B_n = A_n \backslash \bigcup_{i=1}^{n-1} A_i$,
Then $(B_i)$ disjoint, $\bigcup_{i=1}^{+\infty} A_i = \bigcup_{i=1}^{+\infty} B_i$

$$\mu(\bigcup_{i=1}^{+\infty} A_i) = \mu(\bigcup_{i=1}^{+\infty} B_i) = \sum_{i=1}^{+\infty} \mu(B_i) = \sum_{i=1}^{+\infty} \mu(A_i \backslash \bigcup_{j=1} i-1 A_j) \leq \sum_{i=1}^{+\infty} \mu(A_i)$$

② Let $B_1 = A_1, B_2 = A_2 \backslash A_1, \cdots, B_n = A_n \backslash A_{n-1} \cdots$,

$$\begin{aligned}\mu(\bigcup_{i=1}^{+\infty} A_i) = \mu(\bigcup_{i=1}^{+\infty} B_i) = \sum_{i=1}^{+\infty} \mu(B_i)\\ &= \mu(A_1) + \sum_{i=2}^{+\infty} (\mu(A_i) - \mu A_{i-1})\\ &= \lim_{n\to+\infty} (\mu(A_1) + \sum_{i=2}^{n} (\mu(A_i) - \mu A_{i-1}))\\ &= \lim_{n\to+\infty} \mu(A_n)\end{aligned}$$

③

$$\mu(A_1) - \mu(\bigcap_{i=1}^{+\infty} A_i) = \mu(A_1 \backslash \bigcap_{i=1}^{+\infty} A_i) = \mu(\bigcup_{i=1}^{+\infty} A_1 \backslash A_i)$$

Since $A_1 \nearrow$, by $2 := \lim_{n\to+\infty} (\mu(A_1) - \mu(A_i)) = \mu(A_1) - \lim_{n\to+\infty} \mu(A_i)$

**Definition 2.5.2: $\sigma$-finite measure**

Given a measure space $(\Omega, F, \mu)$, $\mu$ is said to be finite if $\mu(\Omega) < +\infty$
$\mu$ is $\sigma$-finite if there exists $(E_i)_{i=1}^{+\infty}$ such that $\bigcup_{i=1}^{+\infty} E_i \in \Omega$ and $\mu(E_i) < +\infty$

**Example 1**

$(\mathbb{R}, \mathcal{B}(\mathbb{R}), m)$ is $\sigma$-finite

**Definition 2.5.3**

if $F \in \mathcal{F}$ is such that $\mu(F) = 0$, then $F$ is called a $\mu$-null set

**Example 2**

$m(\{a\}) = 0$ because

$$m(\{a\}) = m(\bigcap_{n\geq 1}(a - \frac{1}{n}, a]) = (\text{continuous from above}) \lim_{n\to+\infty} m((a - \frac{1}{n}, a]) = 0$$
$$m(Q) = \sum_{q\in Q} m(\{q\}) = 0$$

**Recall:**
Start with $(R, a(\epsilon), m)$ m is content + countably additive
extention: $(R, a(\epsilon), m) \to (R, \sigma(\epsilon), m)$ m is measure

**Theorem 2.5.2: Caratheodory Extention Theorem**

Let F be an algebra on $\Omega$ , $\mu$ be a countably additive content on $(\Omega, F)$,
If $\mu$ is $\sigma$-finite, then $\mu$ extends to a measure on $(\Omega, \sigma(F))$

**Example 3**

Let $\epsilon = \begin{cases} (a, b], & -\infty \leq a < b < +\infty \\ (a, +\infty), & a \in \mathbb{R} \end{cases}$ Let $m_F : a(\epsilon) \to [0, +\infty)$ be a content, such that

$$m_F([a, b]) = F(b) - F(a), m_F((a, +\infty)) = F(+\infty) - F(a)$$

Where F is a right continuous increasing function on $\mathbb{R}$,

$$F(\pm\infty) = \lim_{x\to\pm+\infty} F(x)$$

Then $m_F$ is a countably additive content on $(\mathbb{R}, a(\epsilon))$
By the extension theorem, $m_F$ extends to a measure on $(\mathbb{R}, \sigma(a(\epsilon)) = \sigma(\epsilon) = \mathcal{B}(\mathbb{R}))$

(F(x)=(x) gives Lebesgue)

**Definition 2.5.4: Lebesgue-Stieltjes mesures**

think of $m_F(A) = \int_A \underbrace{dF(x)}_{\text{R-S integral}}$ , F is the distribution function of the measure

Uniqueness(Dynkin)

**Definition 2.5.5: $\pi$ and $\lambda$ system**

Let $C$ be a collection of sets of $\Omega$
$C$ is a $\pi$-system if:

- $\emptyset \in C$
- $\forall A, B \in C, A \cup B \in C$

C is a $\lambda$-system if:

- $\Omega \in C$
- if $A, B \in C,$ and $A \subseteq B,$ then $B \backslash A \in C$
- if $A_1, A_2, \cdots \in C,$ and $A_1 \subseteq A_2 \subseteq \cdots,$ then $\bigcup_{i=1}^{+\infty} A_i \in C$

**Example 4**

$$\epsilon = \begin{cases} (a, b] \\ (a, +\infty) \end{cases} \quad \text{is a } \pi\text{-system}$$

**Exercise:** if $C$ is both a $\pi$-system and a $\lambda$-system, then it is a $\sigma$-algebra

**Lemma 2.5.3: Dynkins Lemma**

Let C be a $\pi$-system, then any $\lambda$-system containing C also contains the $\sigma(C)$
**Hint:** show that any such $\lambda$-system is also a $\pi$-system

**Theorem 2.5.3: Uniqueness Theorem**

Let C be a $\pi$-system, Let $\mu_1, \mu_2$ be two finite measures on $(\Omega, \sigma(C))$
Suppose that $\mu_1(A) = \mu_2(A)$ on $C$, then $\mu_1(A) = \mu_2(A)$ on $\sigma(C)$

Chapter 3

# Partial Differential Equations

## §3.1 Lecture 1 (02-03)Initial and Boundary Conditions

### §3.1.1 What are the PDEs?

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} - u \tag{3.1}$$

$$u = u(x, y, t) \tag{3.2}$$

(3.1) is a second order PDE

### §3.1.2 Classical solutions

To simplify, we consider here equations of one unknown function $u$ of one or more variable
By a solution we mean a sufficiently smooth function $u$ that satisfies the PDE at every point of the domain of its definition($D$).
Sometimes, we will have to impose some boundary conditions on the domain $D$.
$C^n = \{u : D \to R :$ n-times continuous differentiable w.r.t all variables $\}$ $n$ is the order of the equation

**Example 1**

- Heat equation:

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$$
$$u = u(x, t)$$
$$(t, x) \in D \subset R^2$$
$$u(t, x), \frac{\partial u}{\partial t}, \frac{\partial u}{\partial x}, \frac{\partial^2 u}{\partial x^2}, \frac{\partial^2 u}{\partial t^2}$$
$$n = 2$$
$$\frac{\partial^2 u}{\partial x \partial t} = \frac{\partial^2 u}{\partial t \partial x} \in C^2$$
$$u \in C^2(D)$$

- Some Solution:

$$u(t, x) = t + \frac{x^2}{2}, D = R^2$$

$$u(t,x) = \frac{e^{-\frac{x^2}{4t}}}{2\sqrt{\pi t}}, D = \{(t,x) : t > 0\}$$

$$u(t,x) = e^{-t+ix} = e^{-t}cos(x) + ie^{-t}sin(x), D = R^2$$

### §3.1.3 Initial and Boundary Conditions

What is Initial data?
$u(0,x) = u_0(x)$

What are Boundary Conditions? rigurous definition of Boundary

- Dirichlet Boundary Condition: $u(t,x)|_{\partial D} = g(t,x)$
- Neumann Boundary Condition: When the normal derivative of $u$ is specified $\frac{\partial u}{\partial n}$ where $n$ is the normal to the boundary $\partial(D)$

$$\frac{\partial u}{\partial n}(t,x) = g(t,x)$$

- Mixed Boundary Condition: Combination of Dirichlet and Neumann

### §3.1.4 Linear and Nonlinear waves

**Stationary waves**

$$\frac{\partial u}{\partial t} = 0 \tag{3.3}$$

$$u = u(t,x) \tag{3.4}$$

This is a first order linear homogeneuous PDE
Solution of the form $u = u(t) = c, \forall c \in R$
Let's integral (3.3) :

$$0 = \int_0^t \frac{\partial u}{\partial t}(s,x)ds = u(t,x) - u(0,x)$$

Thus if an initial conditional

$$u(0,x) = f(x)$$

$$u(t,x) = f(x), \forall t, x$$

Thus for any given $f \in C^1(R)$, we have a solution of (3.3) and this is a stationary wave
The domain of this solution is $R$ if $f$ is defined everywhere on $R$
Power of domain to change the solution:
Consider the ODE:

$$\frac{\mathrm{d}u}{\mathrm{d}t} = 0, D = (-\infty, 0) \cup (0, \infty)$$

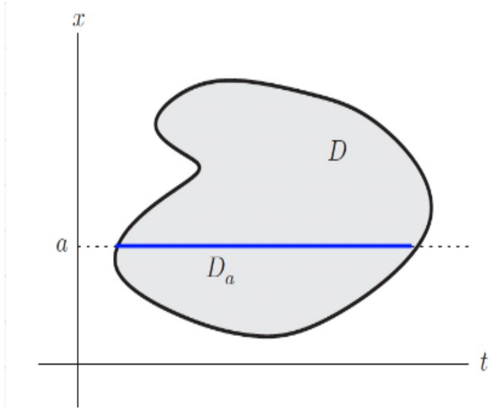
The solution is:

$$u(t) = \begin{cases} c_1, t < 0 \\ c_2, t > 0 \end{cases} \quad \forall c_1, c_2 \in R$$

similarly, for the PDE in example (3.3):

$$u(t,x) = \begin{cases} 0, x > 0 \\ x^2, x \leq 0, t > 0 \\ -x^2, x \leq 0, t < 0 \end{cases}$$

it is a $\mathbb{C}^1$ solution of (3.3) on $\mathbb{R}^2 \backslash \{(0,x) : x \leq 0\}$, so it is not a function of x alone Actually, it is not hard to check that if u is a clasical solution to (3.3), defined on domain $D \subset \mathbb{R}^2$ whose intersection with any horizontal line $D_a = D \cap \{(t,a) : t \in \mathbb{R}\}, \forall a \in \mathbb{R}$ is either empty or a connected interval, then $u(t,x) = f(x)$ is only a function of x.




## §3.2 Lecture 1 (02-05)Transport and Traveling waves

### §3.2.1 uniform transport

Let us consider the linear, homogeneuous first order PDE:

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0 \tag{3.5}$$

$$u = u(0,x) = f(x) \tag{3.6}$$

$$x \in \mathbb{R}, t_0 = 0, t - t_0 = t, c \in \mathbb{R}$$

**use change of variable:**

$$\xi = x - ct$$

$\xi$ is called characteristic variable
let's look for a solution of (3.5) under the form

$$u(t,x) = v(t,\xi) = v(t, x - ct)$$

**use chain rule:**

$$\frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} - c\frac{\partial v}{\partial \xi}$$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial \xi}$$

Thus:

$$0 = \frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \frac{\partial v}{\partial t} - c\frac{\partial v}{\partial \xi} + c\frac{\partial v}{\partial \xi} = \frac{\partial v}{\partial t}$$

**Conclusion:**

$$\frac{\partial v}{\partial t} = 0$$

Solution is constant $v(\xi)$

$$u(t,x) = v(\xi) = v(x - ct)$$

**Proposition 3.2.0**

if u solves (3.5), then u has the form $u(t,x) = v(x - ct)$ where v is a function of one variable that is $\mathbb{C}^1$-smooth.
Acutually, v is satisfying

$$u(0,x) = v(x) = f(x)$$

**Conclusion:**

$$\begin{cases} \frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0 \\ u = u(0,x) = f(x) \end{cases}$$

The solution is

$$u(t,x) = f(x - ct)$$

$f(x - ct)$ is a traveling wave solution moving at velocity $c$:
if c>0, the wave moves to the right
if c<0, the wave moves to the left
if c=0, the wave is stationary

**Example 1**

$$\begin{cases} \frac{\partial u}{\partial t} + 2\frac{\partial u}{\partial x} = 0 \\ u = u(0,x) = \frac{1}{1+x^2} \end{cases}$$

The solution is

$$u(t,x) = \frac{1}{1 + (x - 2t)^2}$$

The lines $x = ct + k$ are called the characteristic lines for the equation (3.5)
on the characteristic lines, the solution is always constant
on geometric interpretation, $u(t,x) = f(x - ct) = f(k)$

Consider the equation:

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} + au = 0 \tag{3.7}$$

where a and c are given numbers.
**Use change of variable:**

$$u(t,x) = v(t,\xi) = v(t, x - ct)$$

we will get by the same computation

$$\frac{\partial v}{\partial t} + av = 0$$

The solution will be

$$v(\xi) = e^{-at} f(\xi)$$
$$u(t,x) = v(t,\xi) = v(x - ct) = e^{-at} f(x - ct)$$

So,

$$\begin{cases} u(t,x) = e^{-at} f(x - ct) \\ u(0,x) = f(x) \end{cases}$$

## §3.2.2 Nonuniform transport

$$\frac{\partial u}{\partial t} + c(x)\frac{\partial u}{\partial x} = 0 \tag{3.8}$$

In the uniform case(c constant), characteristic line is x=ct+k (the solution of ODE below)

$$\frac{\mathrm{d}x}{\mathrm{d}t} = c$$

In the general non-uniform case, we can still define characteristic curves

$$\frac{d}{dt}x(t) = c(x(t))$$

Let's show that the solutions of (3.8) are constants along the characteristic curves (t,x(t))
We need to check that $u(t, x(t)) = h(t)$ is constant of t
compute the derivative of $h(t)$

$$\begin{aligned}\frac{d}{dt}h(t) = \frac{d}{dt}u(t, x(t)) &= \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} \cdot \frac{\mathrm{d}x}{\mathrm{d}t} \\ &= \frac{\partial u}{\partial t} + c(x(t)) \cdot \frac{\partial u}{\partial x} = 0\end{aligned}$$

h(t) and thus u(t,x(t)) is constant along the characteristic curves

**Definition 3.2.1**

The graph of a solution of the ODE

$$\frac{d}{dt}x(t) = c(x(t))$$

is called a characteristic curve for the transport equation (3.8) with speed c(x)

Thus at each point (t,x) the slope of the characteristic curve is c(x) (the wave speed)
Thus, we proved:

**Proposition 3.2.0**

The solutions to the linear transport equation (3.8) are constant along the characteristic curves

In the particular case c(x)=c, the characteristic curves are straight lines of slope c
Moreover,
(3.8) is an autonomous first order ODE which can be "solved" using the method of separation of variables
Assume that c(x)≠0

$$\begin{aligned}\frac{dx(t)}{dt} &= c(x(t)) \\ \frac{dx}{c(X)} &= dt \rightarrow \\ \beta(x) &= \int \frac{dx}{c(x)} = t + k\end{aligned}$$

Thus characteristic curves satisfies the implicit equatioin

$$\beta(x) = t + k$$

We can find x from the equality using $\beta^{-1}$(the inverse of $\beta$)

$$\begin{aligned} \beta(x) &= t + k \\ \beta^{-1}\beta(y) &= y \\ x(t) &= \beta^{-1}(t + k) \end{aligned}$$

**Remark 1**

How to determine where $c(x_0) = 0$?
Solve this ODE

$$\begin{cases} \frac{dx}{dt} = c(x) \\ x(t_0) = x_0 \end{cases}$$

and $x(t) = x_0$ wil be the characteristic curve going through $(t_0, x_0)$

From the proposition, we know that $u(t, x(t))$=constant

$$\begin{aligned} u(t, x(t)) &= v(k) \\ &= v(\beta(x(t)) - t) \end{aligned}$$

where k is the constant determining the characteristic curve (t,x(t))

$$\begin{aligned} u(t, x) &= v(\xi) \\ \text{where } \xi &= \beta(x) - t \end{aligned}$$

which we will call characteristic variable

$$u(t, x) = v(\beta(x) - t) \tag{3.9}$$

This formula way fail at points where c vanishes

$$\begin{aligned} t &= 0 \\ u(0, x) = f(x) &= v(\beta(x) - 0) \\ f(x) &= v(\beta(x)) \\ v(\xi) &= f(\beta^{-1}(\xi)) \\ &= f \cdot \beta^{-1}(\xi) \\ u(t, x) &= f \cdot \beta^{-1}(\beta(x) - t) \end{aligned}$$

This is the formula for the solution of (3.8) in the general case
which is:

$$\begin{cases} \frac{\partial u}{\partial t} + c(x)\frac{\partial u}{\partial x} = 0 \\ u(0, x) = f(x) \end{cases}$$

**Example 2**

Let us apply the method of characteristics to solve the nonuniform transport equation:

$$\frac{\partial u}{\partial t} + \frac{1}{x^2+1}\frac{\partial u}{\partial x} = 0$$

The characteristic curves are the graphs of the solutions of the ODE:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{1}{x^2+1}$$

$$\begin{aligned} \Rightarrow \beta(x) &= \int (x^2+1)dx \\ &= \frac{x^3}{3} + x \\ &= t + k \end{aligned}$$

It is not particularly helpful to find x(t) form this equation.
characteristic variable $\xi$ is given by $\xi = \frac{x^3}{3} + x - t$ and the general solution is of the form

$$u = v(\frac{x^3}{3} + x - t)$$

where v is any $\mathbb{C}^1$ function.

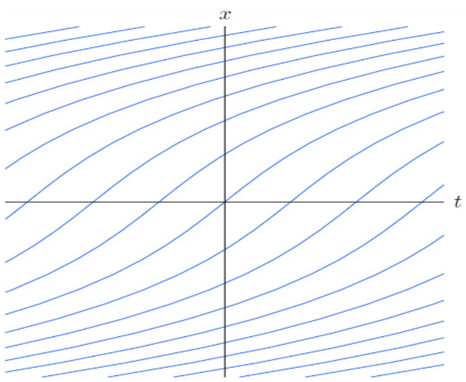
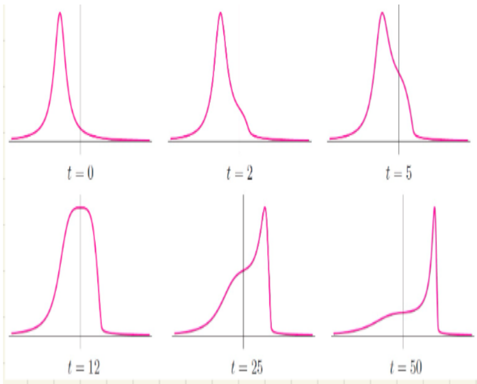



Figure 3.1: Characteristic curves for $u_t + (x^2+1)^{-1}u_x = 0$

In the case of the initial condition:

$$u(0,x) = \frac{1}{1+(x+3)^2}$$

The solution looks as follows:




## §3.3 Lecture 3 (02-10)-Nonlinear transport and shocks

Nonlinear transport equation:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0 \tag{3.10}$$

characteristic equation:

$$\frac{\mathrm{d}x(t)}{\mathrm{dt}} = u(t, x(t)),\ \text{x(t) is the characteristic curve} \tag{3.11}$$

Let's show that u the solution of (3.10) is constant along the characteristic curves x(t) we need to check that $h(t) = u(t, x(t))$ is constant of t

$$\begin{aligned}\frac{d}{dt}h(t) &= \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}\cdot\frac{\mathrm{d}x}{\mathrm{dt}} \\ &= \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x}\cdot u(t, x(t)) \\ &= 0(\text{ by } (3.10))\end{aligned}$$

As u is constant along the characteristic curves, it must be of the form function of characteristic variable:

$$\xi = x - ut$$

$$u(t, x) = f(x - tu(t, x)) \tag{3.12}$$

**Example 1**

Suppose the initial data is:

$$f(\xi) = \alpha\xi + \beta$$

Then (3.12) becomes:

$$u = \alpha(x - tu) + \beta$$

$$u(t, x) = \frac{\alpha x + \beta}{1 + \alpha t} \tag{3.13}$$

As far as $1 + dt \neq 0$, the graoh of the solution is line

If $\alpha > 0$ ,

$$\begin{aligned} u(t,x) &\to 0 \\ t &\to +\infty \end{aligned}$$

If $\alpha < 0$ , then the solution blows up as $t \to -\frac{1}{\alpha}$ since $\dot{x} = u(t, x(t))$
As u(t,x(t))is a constant ,

$$\begin{aligned} u(t, x(t)) &= u(0, x(0)) \\ &= u(0, y) \\ &= f(y) \end{aligned}$$

Our characteristic ODE is:

$$\begin{aligned} \dot{x} &= f(x) \\ \Rightarrow x(t) &= tf(y) + y \\ \Rightarrow u(t, x(t)) &= u(t, tf(y) + y) \\ &= u(0, y) \\ &= f(y), \forall t \\ u(t, tf(y) + y) &= f(y) \forall t, y \\ u(t, x) &= f(y) \\ x &= tf(y) + y \end{aligned}$$

I find $y \in \mathbb{R}$ Suppose let $f(y) = y$

$$\begin{aligned} x = ty + y, y &= \frac{x}{t+1} \\ \Rightarrow u(t,x) = f(y) = y &= \frac{x}{t+1} \\ \Rightarrow u(t,x) &= \frac{x}{t+1} \end{aligned}$$

which agrees with (3.13) for $\alpha = 1, \beta = 0$
There is a problem with this construction that becomes clear when considering the following example:

$$u(0, x) = f(x) = \frac{1}{2}\pi - tan^{-1}(x)$$

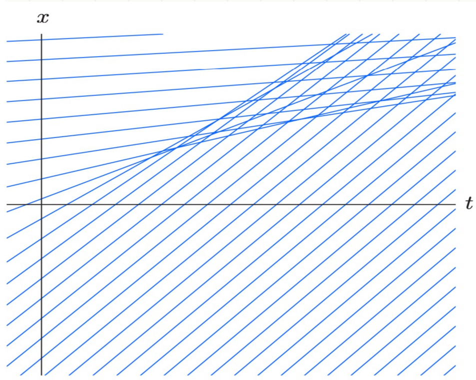



Figure 3.2: Characteristic lines for $u(0, x) = \frac{1}{2}\pi - tan^{-1}(x)$

Two characteristic lines that are not parallel must cross and the value of the solution is supposed to be equal the slope of the characteristic line passing through the point.
At a crossing point, the solution will have to be equal to two different values, one corresoponding to each characteristic line.
While mathematically, such a multiply valued solutions are possible, they are physically untenbale.
One needs to decide which (if any) of the possible values is the physically appropriate.

### §3.3.1 The wave equation:d'Alembert's formula

Let us consider the 1D wave equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \tag{3.14}$$

where c>0 is a given constant
the wave speed, the initial data are of the form:

$$u(0, x) = f(x), \frac{\partial u}{\partial t} = g(x) \tag{3.15}$$

The initial value problem is to find a $\mathbb{C}^2$ function that solves (3.14) and satisfies (3.15)
In this section, we will consider only the 1D case posed on the whole line R
Let us define the wave operator:

$$\Box = \frac{\partial^2}{\partial t^2} - c^2 \frac{\partial^2}{\partial x^2}$$

Thus the wave equation becomes:

$$\Box u = 0$$

In analogy with the elementary factorization

$$t^2 - c^2 x^2 = (t - cx)(t + cx)$$

We can write

$$\Box = (\frac{\partial}{\partial t} - c\frac{\partial}{\partial x})(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x})$$

Now, if

$$(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x})u = 0 \tag{3.16}$$

then u is automatically a solution of the wave equation, because

$$\Box u = (\frac{\partial}{\partial t} - c\frac{\partial}{\partial x})(\frac{\partial}{\partial t} + c\frac{\partial}{\partial x})u = 0$$

Clearly, (3.16) is a transport equation with constant wave speed c, and its solutions are traveling waves with speed c:

$$u(t, x) = p(\xi) = p(x - ct)$$

where p is an arbitrary function.
If $p \in \mathbb{C}^2, u(t, x)$ is a classical solution of the wave equation (3.14).
The factorization of $\Box$ can be written in the reverse order:

$$\Box = (\frac{\partial}{\partial t} + c\frac{\partial}{\partial x})(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x})$$

Solving the transport equation:

$$(\frac{\partial}{\partial t} - c\frac{\partial}{\partial x})u = 0$$

with -c instead of c, we get:

$$u(t, x) = q(x + ct)$$

where q is an arbitrary function.
the solution represent traveling waves moving to the left with constant speed c>0
If $q \in \mathbb{C}^2$, we get a second family of solutions to the wave equation.
Thus there are both left and right traveling-wave solution,
Thus, we have two families of solutions of the wave equation:

$$u(t, x) = p(x - ct)$$
$$u(t, x) = q(x + ct), \forall p, q \in \mathbb{C}^2$$

Any linear combination of these solutions is still a solution of (3.14):

$$u(t, x) = p(x - ct) + q(x + ct), c > 0$$

**Theorem 3.3.1**

Every solution to (3.14) can be written in the form

$$u(t, x) = p(x - ct) + q(x + ct), \forall p, q \in \mathbb{C}^2 \tag{3.17}$$

**Proof:**

$$\xi = x - ct$$
$$\eta = x + ct$$
$$x = \frac{\xi + \eta}{2}, t = \frac{\eta - \xi}{2c}$$
$$u(t,x) = u(\frac{\eta - \xi}{2c}, \frac{\xi + \eta}{2})$$
$$\text{so, } u(t,x) = v(\xi, \eta)$$

Lets find a PED for v:

$$\frac{\partial u}{\partial t} = c(-\frac{\partial v}{\partial \xi} + \frac{\partial v}{\partial \eta})$$
$$\frac{\partial^2 u}{\partial t^2} = c^2(\frac{\partial^2 v}{\partial \xi^2} - 2\frac{\partial^2 v}{\partial \xi \partial \eta} + \frac{\partial^2 v}{\partial \eta^2})$$
$$\frac{\partial^2 u}{\partial x^2} = c^2(\frac{\partial^2 v}{\partial \xi^2} + 2\frac{\partial^2 v}{\partial \xi \partial \eta} + \frac{\partial^2 v}{\partial \eta^2})$$
$$0 = \Box u = \frac{\partial^2 u}{\partial t^2} - c^2\frac{\partial^2 u}{\partial x^2}$$
$$= -4c^2\frac{\partial^2 v}{\partial \xi \partial \eta} = 0$$
$$\Leftrightarrow \frac{\partial^2 v}{\partial \xi \partial \eta} = 0$$
$$\frac{\partial}{\partial \xi}\frac{\partial v}{\partial \eta} v = 0$$
$$\Leftrightarrow \frac{\partial w}{\partial \xi} = 0$$
$$\Leftrightarrow w(\xi, \eta) = w(\eta)$$
$$\Leftrightarrow w = r(\zeta)$$

where $\zeta$ is an arbitary function of $\eta$

$$w = \frac{\partial v}{\partial \eta} = r(\zeta)$$

Integrating we get:

Now we want to solve the initial value problem (3.17):

$$
\begin{aligned}
u(t,x) &= p(x-ct)+q(x+ct)\\
u(0,x) &= p(x)+q(x)=f(x)\\
\frac{\partial u}{\partial t} &= -cp'(x)+cq'(x)=g(x)\\
&\begin{cases} p'(x)+q'(x) & = f(x)\\ -cp'(x)+cq'(x) & = g(x)\end{cases}\\
2p'(x) &= f(x)+\frac{g(x)}{c}\\
p(x) &= \frac{1}{2}f(x)-\frac{1}{2c}\int_0^x g(z)dz+a\\
&\text{where a is an integration constant}\\
q(x) &= f(x)-p(x)\\
&= \frac{1}{2}+\frac{1}{2c}\int_0^x g(z)dz-a\\
u(t,x) &= p(\xi)+q(\eta)\\
&= \frac{f(\xi)+f(\eta)}{2}-\frac{1}{2c}\int_0^\xi g(z)dz+\frac{1}{2c}\int_0^\eta g(z)dz\\
&= \frac{f(\xi)+f(\eta)}{2}+\frac{1}{2c}\int_\eta^\xi g(z)dz
\end{aligned}
$$

**This is called d'Alembert's solution to the IVP** (3.15)

**Theorem 3.3.2: d'Alembert's solution**

The solution of the problem:

$$
\begin{aligned}
\frac{\partial^2 u}{\partial t^2} &= c^2\frac{\partial^2 u}{\partial x^2}\\
u(0,x) &= f(x)\\
\frac{\partial u}{\partial t} &= g(x), x\in\mathbb{R}
\end{aligned}
$$

is given by:

$$
u(t,x)=\frac{f(x+ct)+f(x-ct)}{2}+\frac{1}{2c}\int_{x-ct}^{x+ct} g(z)dz, \forall f\in\mathbb{C}^2, g\in\mathbb{C}^1
$$

Note that this formula provides a classical solution if $f\in\mathbb{C}^2$ and $g\in\mathbb{C}^1$

**External forcing:**
If a homogeneous medium is subject to an external force F, then the system is described as:

$$
\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2}+F(x,t) \tag{3.18}
$$

We apply a generalized version of d'Alembert's technique. To simplify,we first impose homogeneous initial data:

$$\begin{aligned} u(0,x) &= 0 \\ \frac{\partial u}{\partial t} &= 0 \end{aligned}$$

We search for a solution in the form $u(t,x) = v(\xi,\eta)$ and from the chain rule,we get

$$\frac{\partial^2 v}{\partial \xi \partial \eta} = \frac{1}{-4c^2} F(\frac{\eta-\xi}{2c}, \frac{\eta+\xi}{2})$$

Intergrating with respect to $\eta$, we get

$$\frac{\partial v}{\partial \xi} - \frac{\partial v}{\partial \xi} = \frac{1}{-4c^2} \int_\xi^\eta F(\frac{\zeta-\xi}{2c}, \frac{\zeta+\xi}{2}) d\zeta$$

From the identities:

$$\frac{\partial u}{\partial t} = c(-\frac{\partial v}{\partial \xi} + \frac{\partial v}{\partial \eta}), \frac{\partial u}{\partial x} = c(\frac{\partial v}{\partial \xi} + \frac{\partial v}{\partial \eta})$$

We get:

$$\frac{\partial v}{\partial \xi} = -\frac{1}{2c}\frac{\partial u}{\partial t} + \frac{1}{2c}\frac{\partial u}{\partial x}$$

Taking $\xi = \eta$ :

$$\frac{\partial v}{\partial \xi} = -\frac{1}{2c}\frac{\partial u}{\partial t} + \frac{1}{2c}\frac{\partial u}{\partial x} = 0$$

because of homogeneuous initial conditions. Thus (3.18) becomes:

$$\frac{\partial v}{\partial \xi} = \frac{1}{-4c^2} \int_\xi^\eta F(\frac{\zeta-\xi}{2c}, \frac{\zeta+\xi}{2}) d\zeta$$

Now, we integrate this with respect to $\xi$:

$$-v(\xi,\eta) = v(\eta,\eta) - v(\xi,\eta) = \frac{1}{-4c^2} \int_\xi^\eta \int_\xi^\zeta F(\frac{\zeta-\xi}{2c}, \frac{\zeta+\xi}{2}) d\zeta d\xi$$

since $v(\xi,\xi) = 0$, Thus:

$$\begin{aligned} v(\xi,\eta) &= \frac{1}{4c^2} \int_\xi^\eta \int_\phi^\zeta F(\frac{\zeta-\phi}{2c}, \frac{\zeta+\phi}{2}) d\zeta d\phi \\ &= \frac{1}{4c^2} \iint_{T(\xi,\eta)} F(\frac{\zeta-\phi}{2c}, \frac{\zeta+\phi}{2}) d\zeta d\phi \end{aligned}$$

That is the double integral takes place over the triangle

$$T(\xi,\eta) = \{(\phi,\zeta) : \xi \le \phi \le \zeta \le \eta\}$$

Recalling that $\xi = x - ct$ and $\eta = x + ct$ and setting $\phi = y - cs$ and $\zeta = y + cs$, the inequality $\xi \le \phi \le \zeta \le \eta$ becomes $x - ct \le y - cs \le y + cs \le x + ct$,
So T$(\xi, \eta)$ cna be rewritten as $D(t,x) = \{(s,y) | x - c(t-s) \le y \le x + c(t-s), 0 \le s \le t\}$

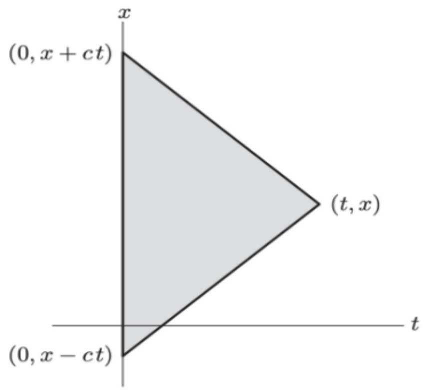



Figure 3.3: The Domain of integration $D(t,x)$

Using the change of variables $\frac{\zeta - \phi}{2c} = s$ and $\frac{\zeta + \phi}{2} = y$ for the double integrals and computing the Jacobian, we get: $\det \begin{pmatrix} \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial s} \\ \frac{\partial \zeta}{\partial y} & \frac{\partial \zeta}{\partial s} \end{pmatrix} = \det \begin{pmatrix} 1 & -c \\ 1 & c \end{pmatrix} = 2c$
Thus,

$$\begin{aligned} u(t,x) &= \frac{1}{2c} \iint_D (t,x) F(s,y) ds dy \\ &= \frac{1}{2c} \int_0^t \int_{x-c(t-s)}^{x+c(t-s)} F(s,y) dy ds \end{aligned}$$

is the formula for the solution to the forced wave equation with homogeneuous initial conditioni.
To solve the case of general initial condition, one needs to take the sum of the solution to the forced equation with homogeneuous initial conditions plus the d'Alembert solution to the unforced equation subject to inhomogeneuous boundary conditions.

## §3.4 Lecture 4 (02-12)

**Theorem 3.4.1**

The solution of the problem:

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} + F(x,t) \\ u(0,x) = f(x) \\ \frac{\partial u}{\partial t} = g(x) \end{cases}$$

is given by

$$u(t,x) = \frac{f(x-ct)+f(x+ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(z)dz + \frac{1}{2c}\int_0^t \int_{x-c(t-s)}^{x+c(t-s)} F(s,y)dyds$$

The triangle D(t,x) is called the domain of dependence of the point (t,x)

**Example 1**

$$\frac{\partial^2 u}{\partial t^2} = c^2\frac{\partial^2 u}{\partial x^2} + sin\omega t sin x, u(0,x) = 0, \frac{\partial u}{\partial t} = 0$$

So:

$$\begin{aligned} F(t,x) &= sin\omega t sin x \\ u(t,x) &= \frac{1}{2c}\int_0^t \int_{x-t+s}^{x+t-s} sin\omega s sin y dyds \\ &= \frac{1}{2}\int_0^t sin\omega s[cos(x-t+s) - cos(x+t-s)]ds \\ &= \begin{cases} \frac{sin\omega t - \omega sint}{1-\omega^2} sinx, \omega \neq 1 \\ \frac{sint - tcost}{2} sinx, \omega = 1 \end{cases} \end{aligned}$$

When $\omega \neq 1$ ,the solution is bounded.
If $\omega \in Q$ ,then the solution u is periodic in time.
If $\omega \in I$ ,then u is quasi-periodic(it never exactly repeats) in time.
When $\omega = 1$ the solution is unbounded.

### §3.4.1 Exercises

① Supposed u(t,x) is defined for all $(t,x) \in R^2$ and solves $\frac{\partial u}{\partial t} + 2u = 0$.
Prove that $\lim_{t\to+\infty} u(t,x) = 0$ for all x
SOL:

$$\begin{aligned} u(0,x) &= f(x) \\ u(t,x) &= e^{-2t}f(x) \\ \lim_{t\to+\infty} u(t,x) &= 0 \end{aligned}$$

② Let u(t,x) solve the initial value problem $\frac{\partial u}{\partial t} + u^c = 0, u(0,x) = f(x)$, where f(x) is a bounded $\mathbb{C}^1$ function of $x \in \mathbb{R}$
(a): show that if $f(x) \geq 0$ for all x, then u(t,x) is defined for all $t > 0$, and $\lim_{t\to+\infty} u(t,x) = 0$
(b): On the other hand , if $f(x) < 0$ , then the solution u(t,x) is not defined for all t>0, but in fact, $\lim_{t\to r^-} u(t,x) = -\infty$, for some $0 < r < \infty$ Given x , what is the corresoponding value of r?
(c):Given f(x) as in part (b), what is the longest time interval $0 < t < t_\star$ on which u(t,x) is defined for all $x \in \mathbb{R}$?

SOL:

$$\frac{\partial u}{\partial t} = -u^2, u(0,x) = f(x)$$
$$\frac{du}{u^2} = -dt \Rightarrow -\frac{1}{u} = -t - c(x)$$
$$\frac{1}{u} = t + c(x) \Rightarrow u(t,x) = \frac{1}{t + c(x)}$$
$$u(0,x) = f(x) = \frac{1}{c(x)}$$
$$u(t,x) = \frac{1}{t + \frac{1}{f(x)}} = \frac{f(x)}{tf(x)+1}$$
$$(a) : f(x) \geq 0 \Rightarrow u(t,x) \text{ is defined for any } t \geq 0 \text{ and } \lim_{t\to+\infty} u(t,x) = 0$$
$$(b) : f(x) < 0 \Rightarrow \tau(x) = \frac{1}{f(x)} \Rightarrow u(t,x) = \frac{1}{t + \tau(x)}$$
$$(c) : If f(x) < 0, [0, \tau(x))$$
$$t_\star = \inf\{\tau(x) : x \in \mathbb{R}\} = inf_{x\in\mathbb{R}} - \frac{1}{f(x)} = -\frac{1}{inf_{x\in\mathbb{R}} f(x)}$$

So we can have $f(x) = tan^{-1}(x)$

③ Solve the following initial value problem and graph the solution at times t=1,2,and 3:
(a) $u_t - 3u_x = 0, u(0,x) = e^{-x^2}$
Recall:

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0$$
$$\text{We knnow that } u(t,x) = v(\xi) = (x - ct)$$
$$\forall v \in \mathbb{C}^1$$
$$u(0,x) = f(x) = v(x)$$
$$\text{Thus } u(t,x) = f(x - ct)$$

**In our case:**

$$u(t,x) = f(x + 3t) = e^{-(x+3t)^2}$$

④ Graph some of the characteristic lines for the following equations and write down a formula for the general solution:
(a)$u_t - 3u_x = 0$,
SOL: The characteristic lines of the equation $\frac{\partial u}{\partial t} + c(x)\frac{\partial u}{\partial x} = 0$ are the solution of the ODE:

$$\frac{\mathrm{d}x}{\mathrm{d}t} = c(x)$$

In our case $\frac{\mathrm{d}x}{\mathrm{d}t} = -3 \to x(t) = -3t + k$

⑤ (a)Prove that if the initial data is bounded,$|f(x)| \leq M$ for all $x \in \mathbb{R}$,then the solution to the damped transport equation $\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} + au = 0$ with a>0 satisfies

$u(t,x) \to 0$ as $t \to \infty$
(b) Find a solution to it that is defined for all (t,x) but does not satisfy $u(t,x) \to 0$ as $t \to +\infty$
SOL: Recall the solution to it with initial condition u(0,x)=f(x) is given by

$$\begin{aligned} u(t,x) &= f(x-ct)e^{-at} \\ |u(t,x)| &= |f(x-ct)|e^{-at} \leq Me^{-at} \\ \text{(b) } f(x) &= e^{Ax} \\ u(t,x) &= e^{A(x-ct)-at} = e^{-(Ac+a)t+Ax} \\ \text{Choose } A &\leq -\frac{a}{c} \end{aligned}$$

6 (a) Write down a formula for the general solution to the nonlinear partial differential equation $u_t + u_x + u^2 = 0$.
(b) Show that if the initial data is positive and bounded, $0 \leq u(0,x) = f(x) \leq M$, then the solution exists for all $t > 0$, and $u(t,x) \to 0$ as $t \to \infty$.
(c) On the other hand, if the initial data is negative somewhere, so $f(x) < 0$ at some $x \in \mathbb{R}$, then the solution blows up in finite time: $\lim_{t\to\tau^-} u(t,y) = -\infty$ for some $\tau > 0$ and some $y \in \mathbb{R}$.
(d) Find a formula for the earliest blow-up time $\tau_* > 0$.
SOL: Writing u(t,x) in terms of the characteristic variable $\xi = x - t$ we get:

$$u(t,x) = v(t,\xi) = u(t,\xi + t)$$

Let us find the equation for v:

$$\begin{aligned} \frac{\partial v}{\partial t} &= \frac{\partial u}{\partial t} + \frac{\partial u}{\partial x} = -u^2 = -v^2 \\ \Rightarrow \frac{dv}{v^2} &= -dt \Rightarrow -\frac{1}{v} = -t - c(\xi) \\ \Rightarrow v(t,\xi) &= \frac{1}{t + c(\xi)} \\ \Rightarrow u(t,x) &= \frac{1}{t + c(x-t)} \end{aligned}$$

Where c is an arbitrary $\mathbb{C}^1$ function.
(b): if there is an initial condition $u(0,x) = f(X)$, then we can determinec(x):

$$\begin{aligned} u(0,x) = f(x) &= \frac{1}{c(x)} \\ \Rightarrow u(t,x) &= \frac{1}{t + \frac{1}{f(x-t)}} = \frac{f(x-t)}{tf(x-t)+1} \end{aligned}$$

If $0 \leq f(x)$, then $tf(x-t) + 1 \neq 0$,so u(t,x) id defined for any t>0.
The function $\frac{s}{ts+1}$ is increasing in s
Therefore:

$$0 \leq u(t,x) \leq \frac{M}{Mt+1} \underbrace{\to}_{(\text{as } t\to\infty)} 0 \text{ as } f(x) \leq M$$

(c): The solution formula implies that, if $f(x) < 0$ then

$$\begin{aligned} &u(t, x_\star) \to -\infty \\ &\text{as } t \to \tau = -\frac{1}{f(x)} \\ &where x_\star = x - \frac{1}{f(x)} \end{aligned}$$

(d):

$$\tau_* = -\frac{1}{\inf_{x\in\mathbb{R}} f(x)}$$

Chapter 4

# Algorithm

## §4.1 Lecture 1 (02-03)Asymptotic bounds and big O

**Definition 4.1.1: big-Theta notation(=)**

Let $f, g : N \to R^+$, we say that $f(n) = \Theta(g(n))$ if there $\exists\ a, b > 0$ and $n_0 \in \mathbb{N}$such that $a \cdot g(n) \leq f(n) \leq b \cdot g(n)$ for all $n > n_0$.

Intuitively, $f(n) = \Theta(g(n))$ means that $f(n)$ grows as fast as $g(n)$ up to a constant when $n$ tends to infinity.

Examples: $3n = \Theta(n)$, $2n^2 + 7n = \Theta(n^2)$, $\ln n^2 = \Theta(\log n)$

**Definition 4.1.2: big-O notation(≦)**

Let $f, g : N \to R^+$, we say that $f(n) = O(g(n))$ if $\exists\ n_0 \in \mathbb{N}$ and $b > 0$ such that $f(n) \leq b \cdot g(n)$ for all $n > n_0$.
$\lim_{n\to\infty} sup \frac{f(n)}{g(n)} < +\infty$

**Definition 4.1.3: big-Omega notation(≥)**

Let $f, g : N \to R^+$, we say that $f(n) = \Omega(g(n))$ if $\exists\ n_0 \in \mathbb{N}$ and $a > 0$ such that $a \cdot g(n) \leq f(n)$ for all $n > n_0$.
$\lim_{n\to\infty} inf \frac{f(n)}{g(n)} > 0$

**Remark 1**

$$f(n) = \Theta(g(n)) \Leftrightarrow f(n) = O(g(n)) \text{ and } f(n) = \Omega(g(n))$$
$$f(n) = O(g(n)) \Leftrightarrow g(n) = \Omega(f(n))$$

**Definition 4.1.4: ittle-o notation(<)**

We write $f(n) = o(g(n))$ if for any $b > 0$ there $\exists\ n_0 \in \mathbb{N}$ such that $f(n) < b \cdot g(n)$ for all $n > n_0$.

**Definition 4.1.5: little-omega notation(>)**

We write $f(n) = \omega(g(n))$ if for any $a > 0$ there $\exists n_0 \in \mathbb{N}$ such that $a \cdot g(n) < f(n)$ for all $n > n_0$.

**Remark 2**

$f(n) = o(g(n)) \Rightarrow f(n) = O(g(n))$ and $f(n) \neq \Theta(g(n))$
$f(n) = \omega(g(n)) \Rightarrow f(n) = \Omega(g(n))$ and $f(n) \neq \Theta(g(n))$
$f(n) = \omega(g(n)) \Leftrightarrow g(n) = \omega(f(n))$

$$n^{1+e}, nlog(n), e \to 0$$

**Theorem 4.1.1**

Any Algorithm has to check at least log(n+1) entries of the sorted array for searching an element in the worst case

## §4.2 Recitation 1 (02-07)–Problem Solving

| Notation | Bounding Conditions | Relation | Type |
|---|---|---|---|
| $T(n) = O(f(n))$ | $\exists c > 0, \exists n_0, \forall n \geq n_0, T(n) \leq cf(n)$ | $\leq$ | $O$ |
| $T(n) = \Omega(f(n))$ | $\exists c > 0, \exists n_0, \forall n \geq n_0, T(n) \geq cf(n)$ | $\geq$ | $\Omega$ |
| $T(n) = \Theta(f(n))$ | $\exists c_1, c_2 > 0, \exists n_0, \forall n \geq n_0, c_1 f(n) \leq T(n) \leq c_2 f(n)$ | $=$ | $\Theta$ |
| $T(n) = o(f(n))$ | $\forall c > 0, \exists n_0, \forall n \geq n_0, T(n) < cf(n)$ | $<$ | $o$ |
| $T(n) = \omega(f(n))$ | $\forall c > 0, \exists n_0, \forall n \geq n_0, T(n) > cf(n)$ | $>$ | $\omega$ |

**Possible useful formula:**

$$\int_1^n x^k dx = \frac{n^{k+1}}{k+1} - \frac{1}{k+1}$$

1. Multiplication Rule. Prove that for $h(n) = \Omega(1)$, if $f(n) = \Theta(g(n))$ then $f(n) \cdot h(n) = \Theta(g(n) \cdot h(n))$ (The same also holds for $O, \Omega, \omega, \omega$)
   **Solution:**

   Since $f(n) = \Theta(g(n)), \exists c_1, c_2 > 0, \exists n_0$ such that
   $c_1 g(n) \leq f(n) \leq c_2 g(n), \forall n \geq n_0$
   Since $h(n) = \Omega(1)$, there exist constants $c_3 > 0$ and $n_1$ such that
   $h(n) \geq c_3, \forall n \geq n_1$
   Combining the two, we have:
   $c_1 g(n)h(n) \leq f(n)h(n) \leq c_2 g(n)h(n)$
   Therefore $f(n)h(n) = \Theta(g(n)h(n))$

2. Addition Rule.Let $f_1, ..., f_k : N \to R_{\geq 0}$ for a fixed number $k$ and $f(n) = \sum_{i=1}^{k} f_i(n)$ .Prove that if $f_i(n) = O(f_1(n))$ for all $i \in \{1, \ldots, k\}$ ,then $f(n) = \Theta(f_1(n))$ .Here

$f_1$ can be replaced with any one of $f_1, ..., f_k$
**Solution:**

Since $f_i(n) = O(f_1(n)),$ there exist constants $c_i$ and $n_0$ such that
$f_i(n) \leq c_i f_1(n), \forall n \geq n_0$
Therefore,
$\sum_{i=1}^{k} f_i(n) \leq \sum_{i=1}^{k} c_i f_1(n) = C f_1(n)$ where $C = \sum_{i=1}^{k} c_i$

Additionally, there exists at least one $f_j(n) = \Omega(f_1(n)),$ so $\sum_{i=1}^{k} f_i(n) = \Theta(f_1(n))$

3. Let $f$ and $g$ be non-negative functions. Define $T(n) = max(f(n), g(n))$ .Prove that $T(n) = \Theta(f(n) + g(n))$
**Solution:**

Since $max(f(n), g(n)) \leq f(n) + g(n),$ we have $T(n) = O(f(n) + g(n))$
Also, $f(n) + g(n) \leq 2max(f(n), g(n)),$ so $T(n) = \Omega(f(n) + g(n))$
Therefore, $T(n) = \Theta(f(n) + g(n))$

4. Let $f$ and $g$ be non-decreasing real-valued functions defined on positive integers and $f(1) \geq 1$ and $g(1) \geq 1$ .Prove or disprove that $f(n) = O(g(n)) \Rightarrow 2^{f(n)} = O(2^{g(n)})$
**Solution:**

Since $f(n) = O(g(n)),$ there exists a constant $c > 0$ and $n_0$ such that
$f(n) \leq c \cdot g(n), \forall n \geq n_0$
Therefore,
$2^{f(n)} \leq 2^{c \cdot g(n)}$
Since $2^c$ is constant, we conclude that $2^{f(n)} = O(2^{g(n)})$

5. Prove that $log(n!) = \Theta(nlogn)$ .Hint:To show an upper bound,compare $n!$ with $n^n$ .To show a lower bound, compare it with $\left(\frac{n}{2}\right)^{\frac{n}{2}}$
**Solution:**

We know that:
$log(n!) = log(1) + log(2) + \ldots + log(n)$
Since $log(n)$ is increasing, the sum can be bounded by:
$log(n!) \leq \sum_{k=1}^{n} log(n) = nlog(n)$
From the bound, we can use the fact that $log(k) \geq log(\frac{n}{2})$ for all $k \geq 2$
Hence, we can sum the lower half of the terms to get:
$log(n!) \geq \sum_{k=\frac{n}{2}}^{n} log(k) \geq \sum_{k=\frac{n}{2}}^{n} log(\frac{n}{2}) = \frac{n}{2} log(\frac{n}{2})$

6. Show that $c^n = o(n!)$ for any constant $C$
**Solution:**

We need to prove that
$$\lim_{n\to\infty} \frac{c^n}{n!} = 0$$
Using the recursive relation:
$$\frac{c^n}{n!} = \frac{c^{n-1}}{(n-1)!} \times \frac{c}{n}$$
As $n \to \infty, \dfrac{c}{n} \to 0,$ Thus,
$$\lim_{n\to\infty} \frac{c^n}{n!} = 0$$

7. Show that $c^{n^{1.001}} = \omega(n!)$ for any constant $C$ .Hint: For any fixed numbers $p, q > 0$ ,we have $n^p = \omega(log^q n)$
**Solution:**

We need to prove that
$$\lim_{n\to\infty} \frac{c^{n^{1.001}}}{n!} = \infty$$
Taking the algorithm, we have:
$$log(\frac{c^{n^{1.001}}}{n!}) = n^{1.001} log(c) - log(n!)$$
Since $log(n!) \approxeq nlogn,$ and $n^1.001$ grows faster than $nlogn,$ we conclude that
$$\lim_{n\to\infty} log(\frac{c^{n^{1.001}}}{n!}) = \infty$$
So, $\lim_{n\to\infty} \dfrac{c^{n^{1.001}}}{n!} = \infty$
Therefore, $c^{n^{1.001}} = \omega(n!)$

8. Show $\sum_{i=1}^{n} i^k = \Theta(n^{k+1})$ .Hint: Show that $n^{k+1} - (n-1)^{k+1} = \Theta(n^k)$
**Solution:**

Upperbound: Since $i^k \leq n^k$, we have
$$\sum_{i=1}^{n} i^k \leq \sum_{i=1}^{n} n^k = n^{k+1}$$
Lowerbound: Using the integral approximation
$$\sum_{i=1}^{n} i^k \geq \int_1^n x^k dx = \frac{n^{k+1}}{k+1} - \frac{1}{k+1}$$
Therefore,
$$\sum_{i=1}^{n} i^k = \Theta(n^{k+1})$$

9. Search for the peak. Suppose we have an array $A[1...n]$ and we know that there exists some k (value unknown) such that $A[1] < A[2] < ... < A[k]$ and $A[k] > A[k+1] > \ldots > A[n]$ .We call $A[k]$ the peak of $A$ .Design an algorithm to find the peak and analyze it in $O(logn)$ time.
**Solution:**

    Use Binary Search Algorithm:
    1.Set $l = 1, r = n$
    2.While $l < r$ do
    3.Set $m = \left\lfloor \dfrac{l+r}{2} \right\rfloor$
    4.If $A[m] < A[m+1]$ then $l = m+1$
    5.Else $r = m$
    6.Return $l$
    Analysis:
    Let $T(n)$ be the time complexity of the algorithm
    Let $n_0 = 1$
    Let $c = 1$
    Let $T(n) = T\left(\dfrac{n}{2}\right) + c$
    Let $T(n) = O(logn)$

10. 3SUM in a sorted array.Let $A[1 \ldots n]$ be a sorted array. We want to see $A[i] + A[j] + A[k] = x$ for some different $i, j, k$ ,where $x$ is a given number. Design an algorithm $3SUM(n, a, x)$ which can return $(i, j, k)$ or 0 if no such three elements exist.How fast can this problem be solved?

**Solution:**

fix one element A[i], then use two pointers on the remaining array to find $A[j] + A[k] = x - A[i]$
The time complexity is $O(n^2)$
**Algorithm 3SUM(n,a,x):**
Input: Array A of n intergers, integer x
Output: indices i,j,k such that A[i]+A[j]+A[k]=x or return "No solution" if no such triplet exists
for i=0 to n-1 do
 //Avoid duplicates by skipping repeated elements
 if i>0 and A[i]==A[i-1]
 continue
 left=i+1,right=n-1
 while left<right do
 sum=A[i]+A[left]+A[right]
 if sum==x
 return i,left,right //Found the triplet
 if sum<x
 left=left+1 //increase the sum by moving left pointer to the right
 else
 right=right-1 //decrease the sum by moving right pointer to the left
return "No solution" //no such triplet exists